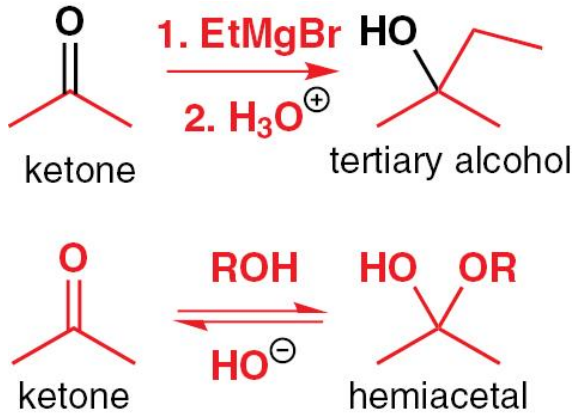
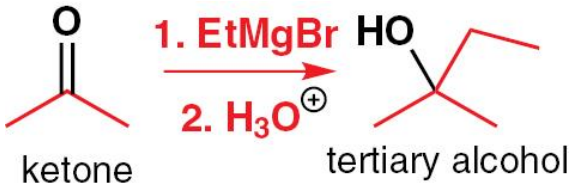
Nucleophilic ***Substitution*** at the Carbonyl Group

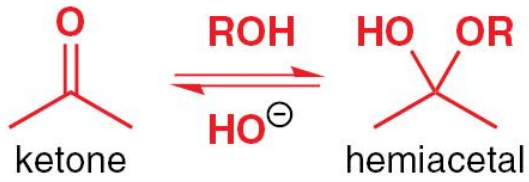
- Nucleophilic attack followed by loss of leaving group
- What makes a good nucleophile
- What makes a good leaving group
- There is always a tetrahedral intermediate
- How to make acid derivatives
- Reactivity of acid derivatives
- How to make ketones from acids
- How to reduce acids to alcohols

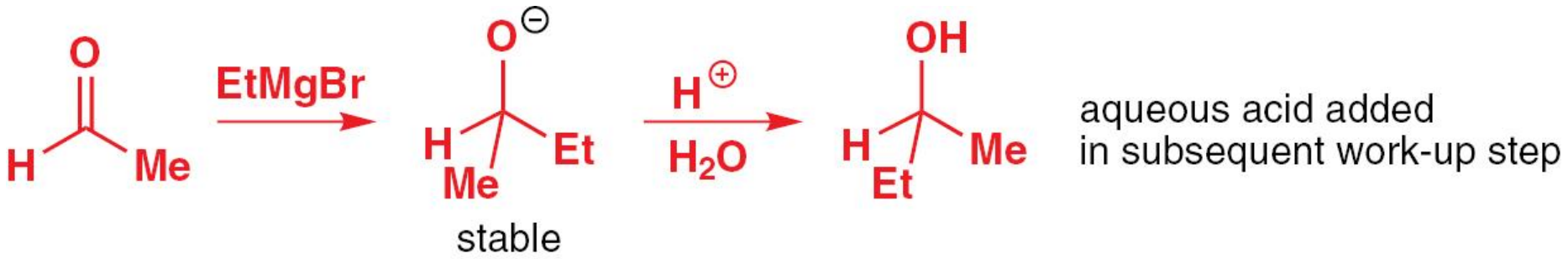
Irreversible and reversible additions to a carbonyl group



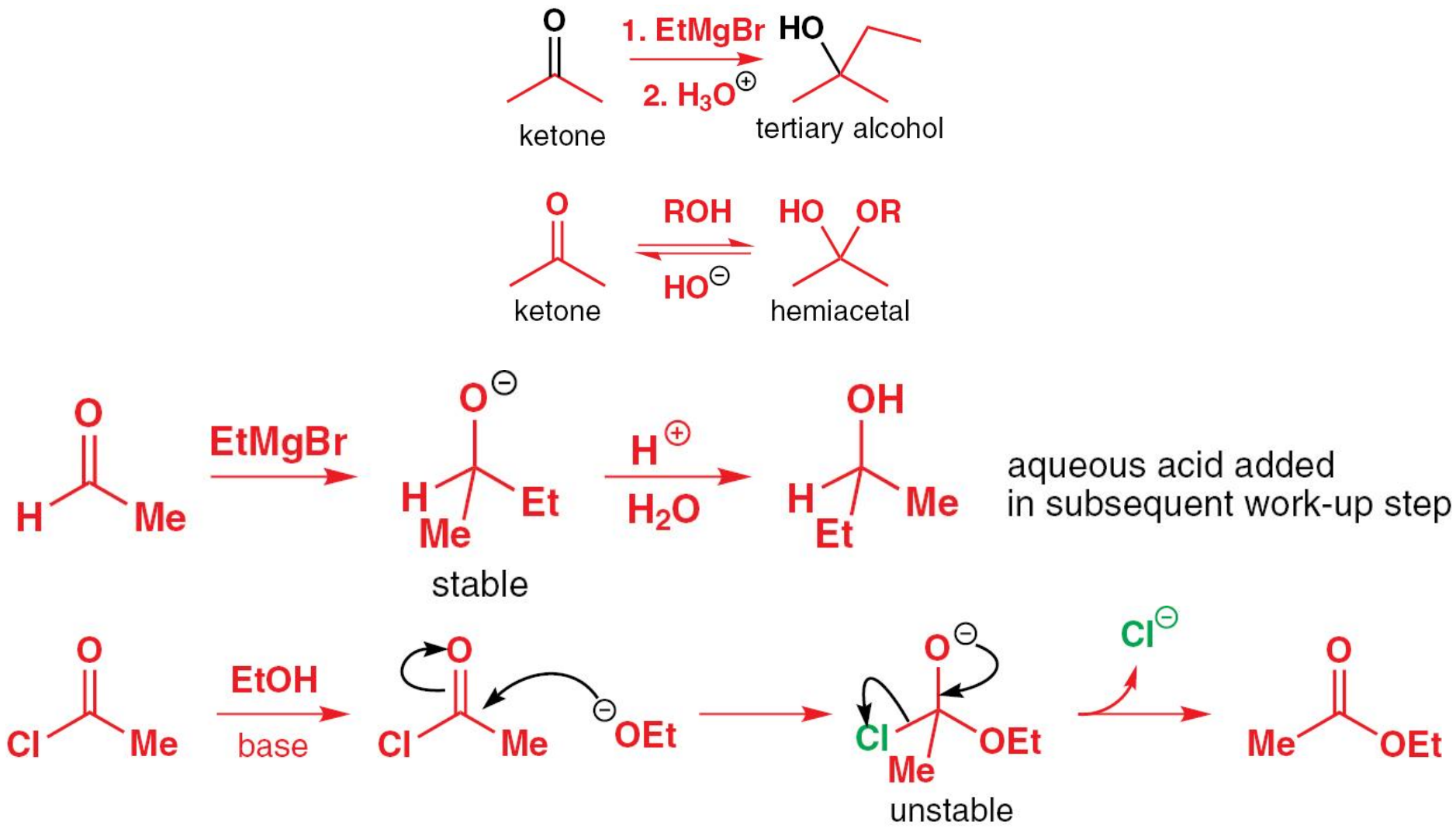
Irreversible and reversible additions to a carbonyl group



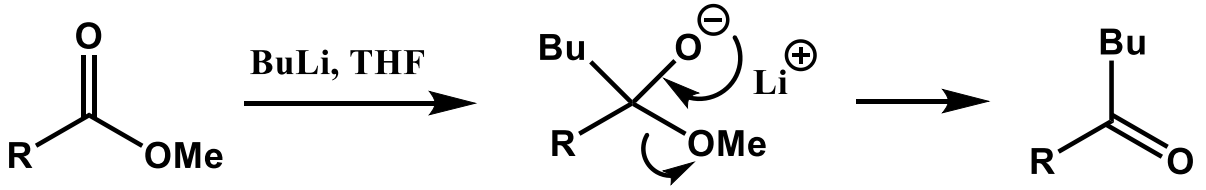




Irreversible and reversible additions to a carbonyl group



Tetrahedral Intermediates

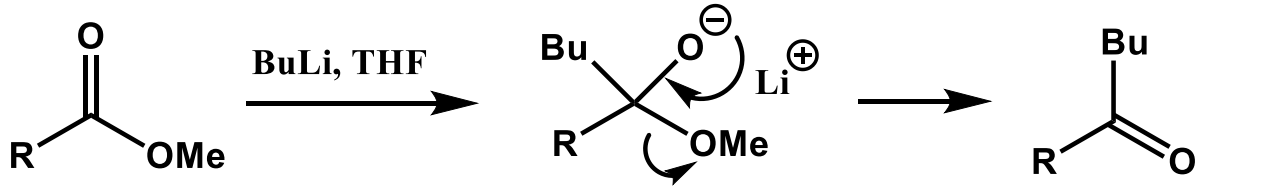


TETRAHEDRAL INTERMEDIATE

$MeO^{\ominus}$

$MeO^{\ominus}$ **is the leaving group**

Tetrahedral Intermediates



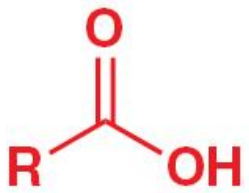
TETRAHEDRAL INTERMEDIATE

$MeO^{\ominus}$

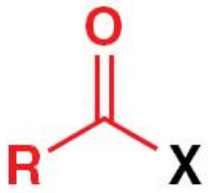
$MeO^{\ominus}$ is the leaving group

If the carbonyl compound contains a potential leaving group (such as RO^-) then the tetrahedral intermediate can fragment with loss of the leaving group

Carboxylic acid derivatives

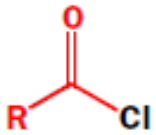


carboxylic acid

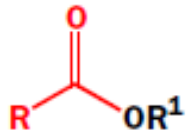


carboxylic acid derivative

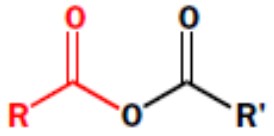
Carboxylic acid derivatives



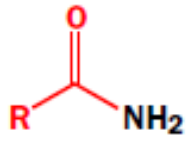
acid chlorides (acyl chlorides)[a]



esters



acid anhydrides



amides

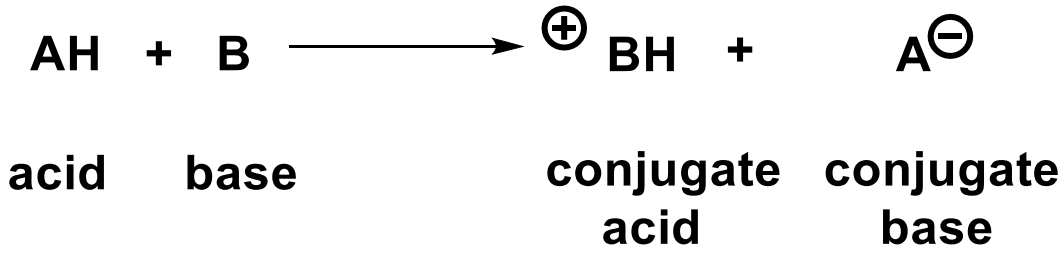
[a]We shall use these two terms interchangeably.

Leaving Groups

A good leaving group can be defined as “the conjugate base of a strong acid

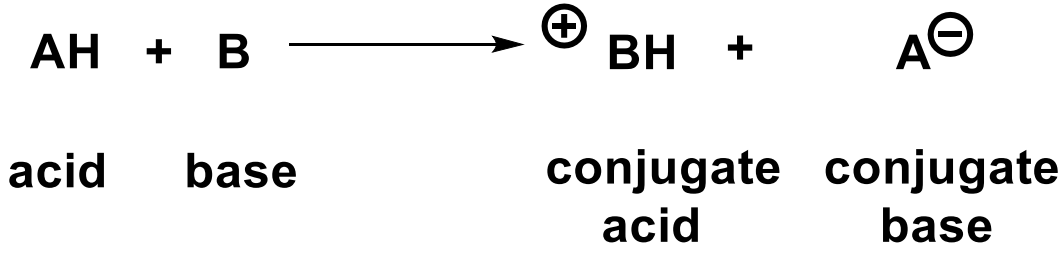
Leaving Groups

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Leaving Groups

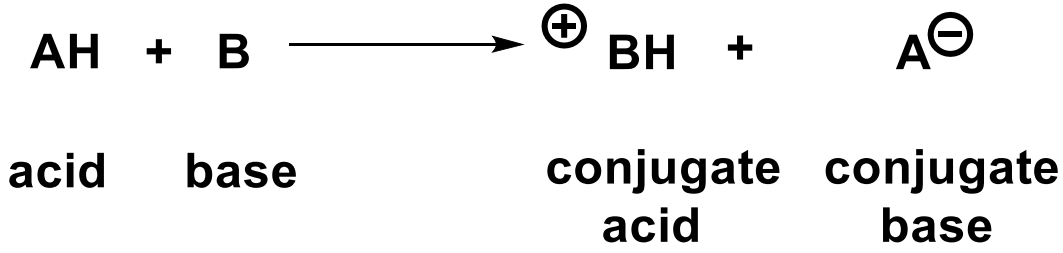
A good leaving group can be defined as "the conjugate base of a strong acid



Acid	p*K*a
MeH	50
NH_3	35
ROH	16
ROH_2^+	-5
ArOH	10
RCOOH	5
H_2O	15,7
H_3O^+	-1,7
HCl	-7

Leaving Groups

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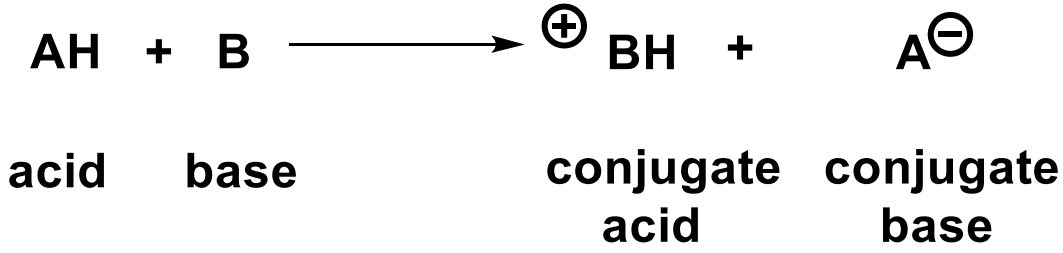
$^{-}$OH is not a good leaving group

H_2O is a good leaving group

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Leaving Groups

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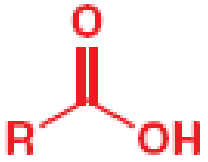
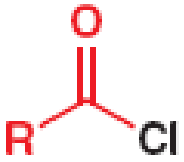
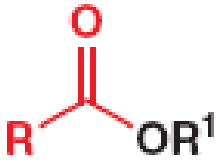
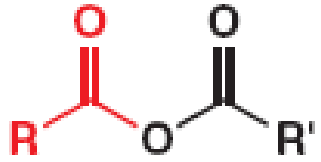
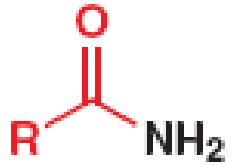
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$^-$OH is not a good leaving group

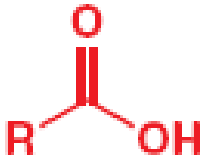
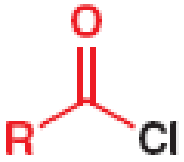
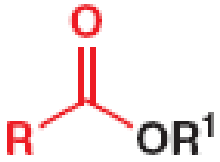
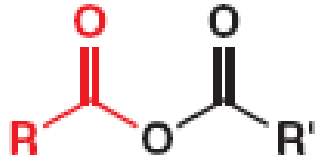
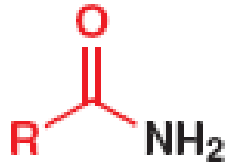
H_2O is a good leaving group

The lowest the p*K*a, the better the leaving group

Carboxylic acid derivatives: Which one has the best leaving group?

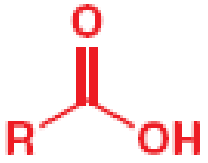
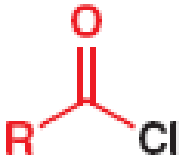
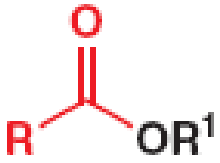
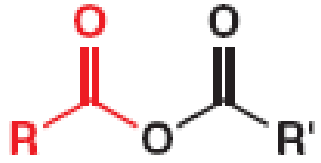
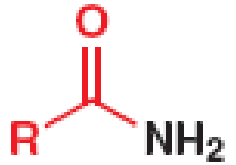
Carboxylic acid	Derivative of RCO_2H			
	acid chloride or acyl chloride*	ester	acid anhydride	amide
				

Carboxylic acid derivatives: Which one has the best leaving group?

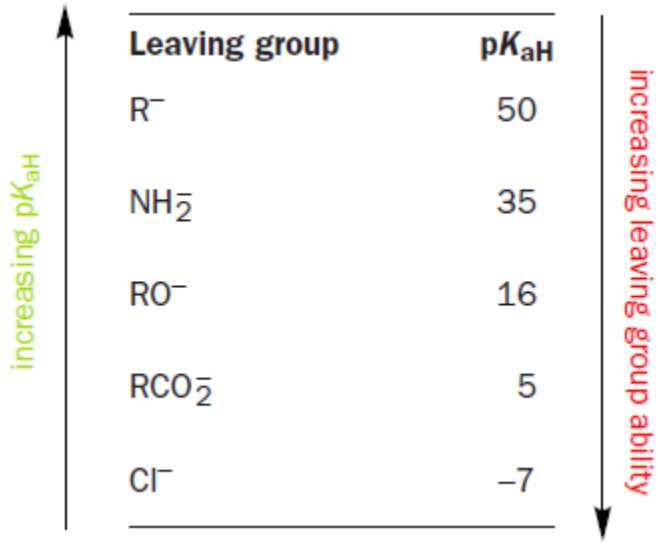
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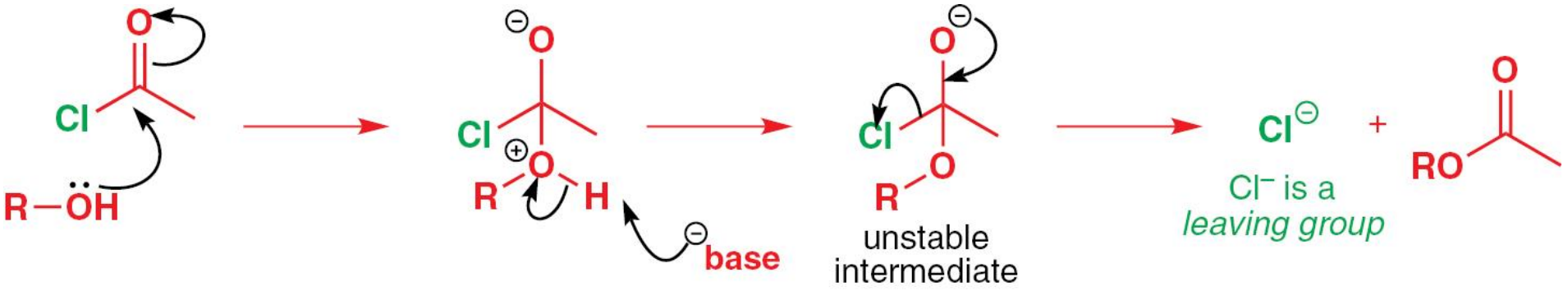
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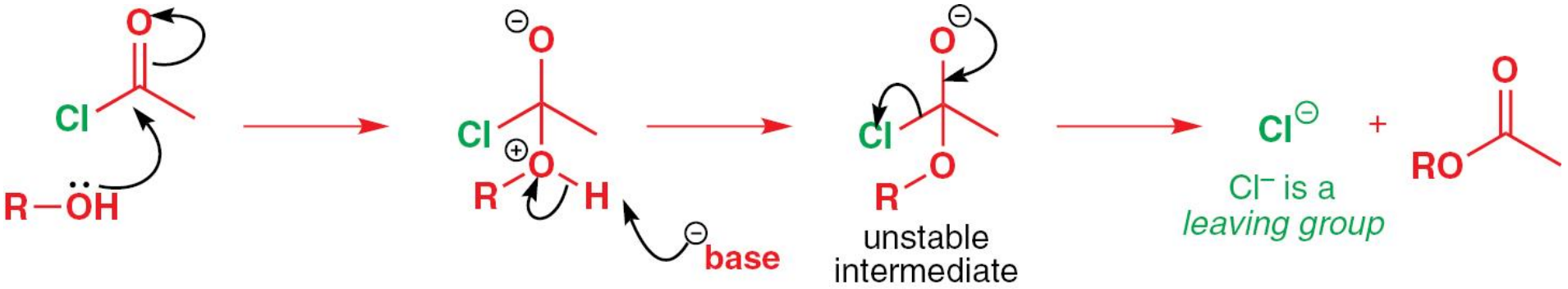


Making esters from acyl chlorides

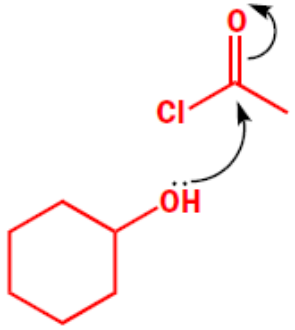


Esters are formed by nucleophilic substitution of alcohols

Making esters from acyl chlorides

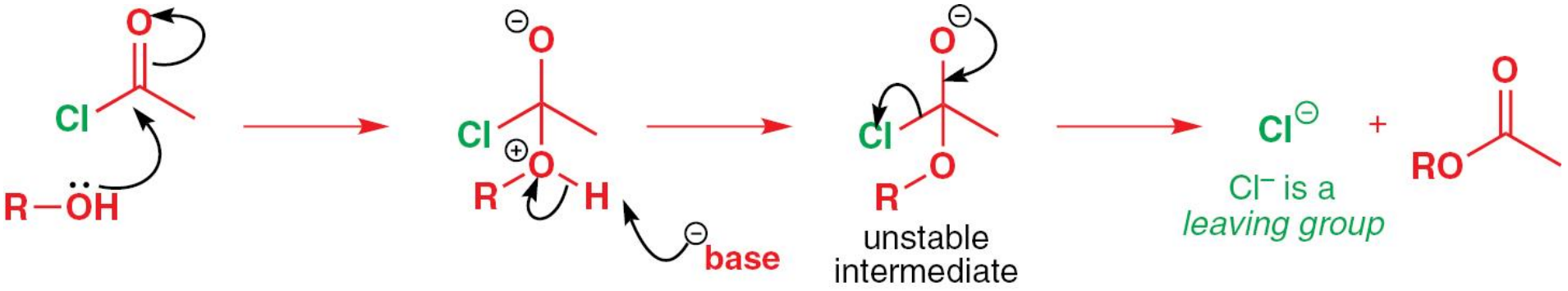


Example

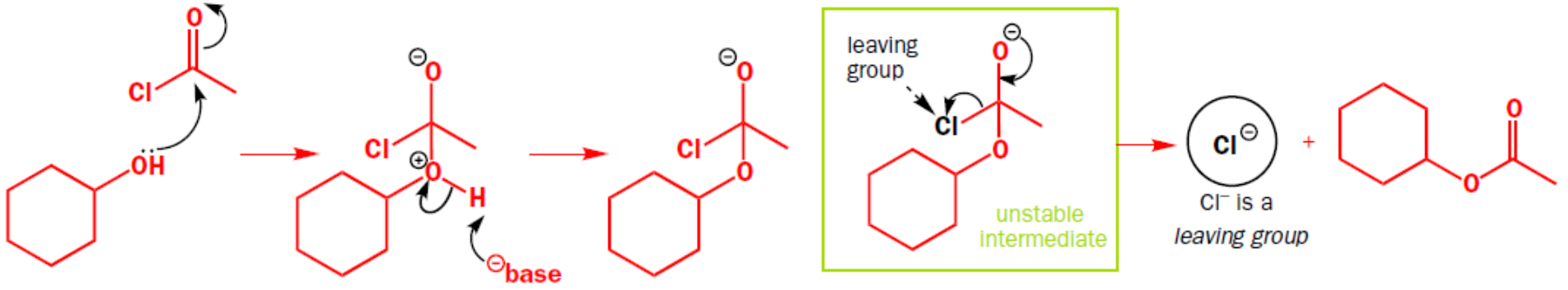


Esters are formed by nucleophilic substitution of alcohols

Making esters from acyl chlorides

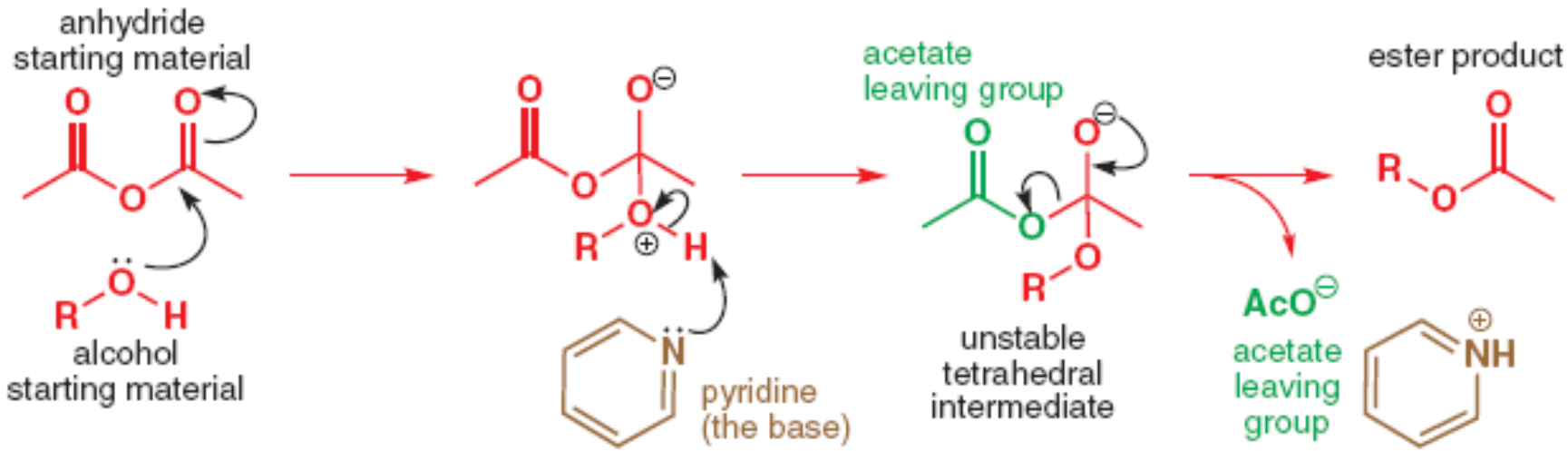


Example

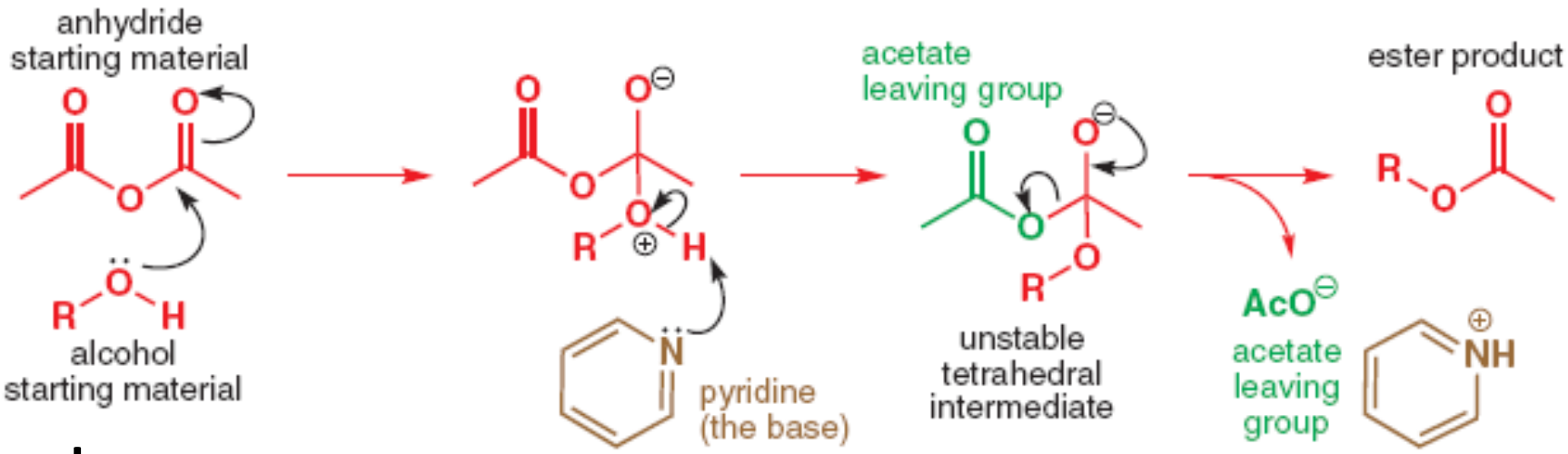


Esters are formed by nucleophilic substitution of alcohols

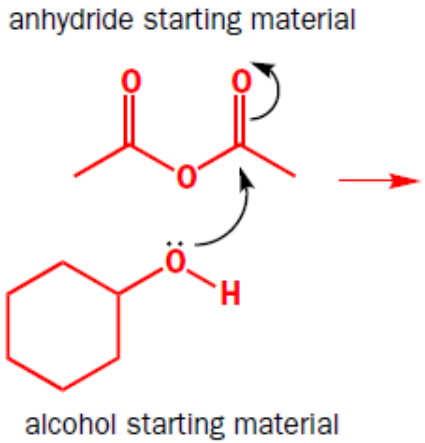
We can form esters also from acid anhydrides



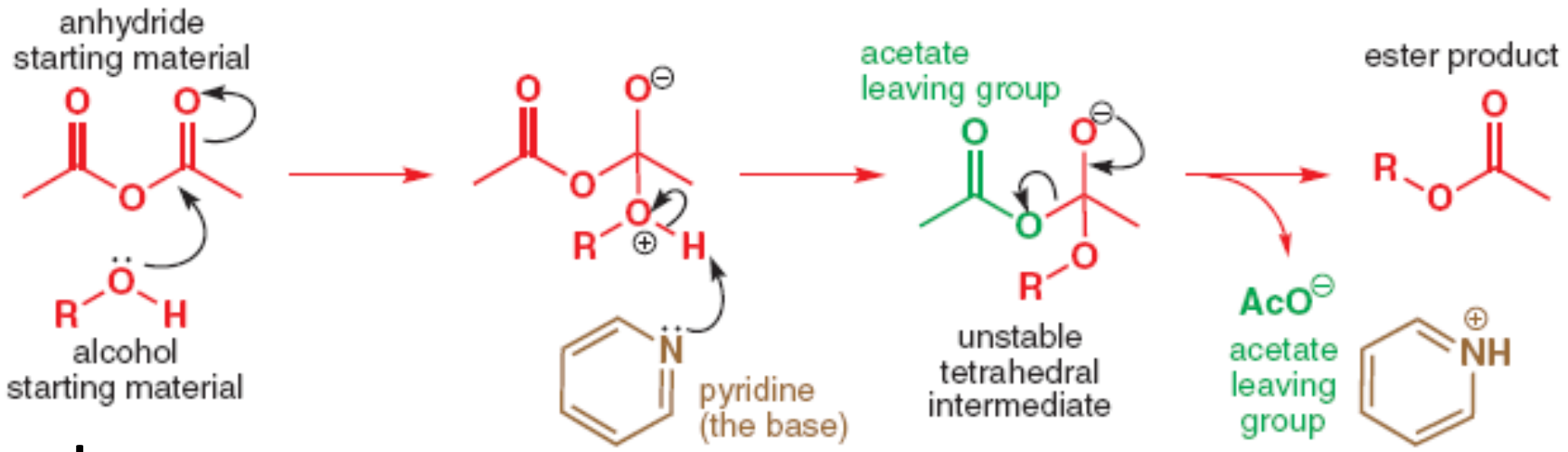
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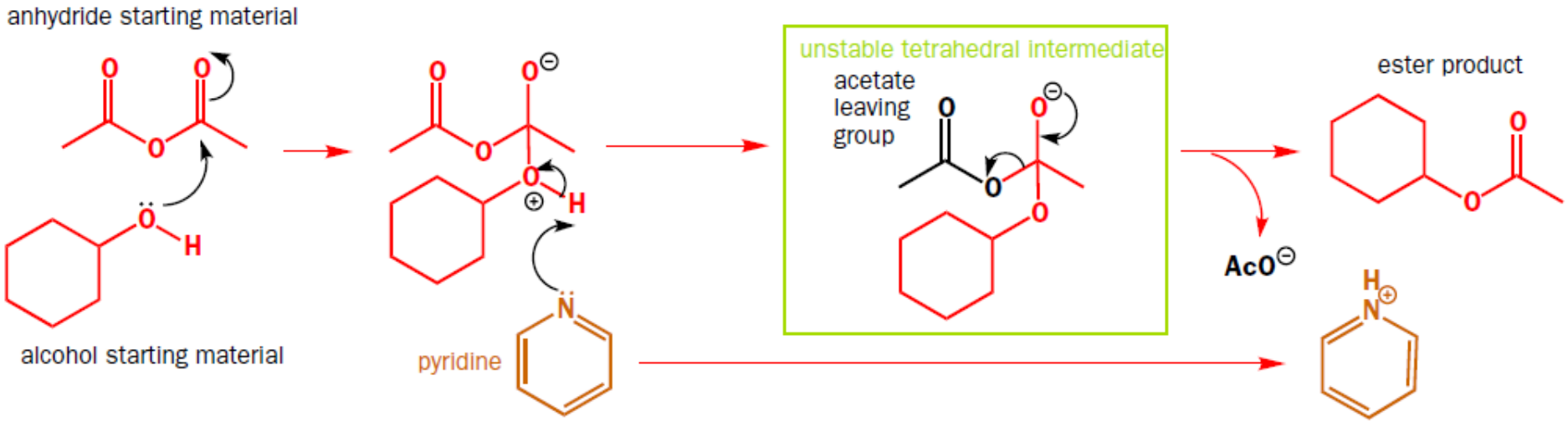
Example



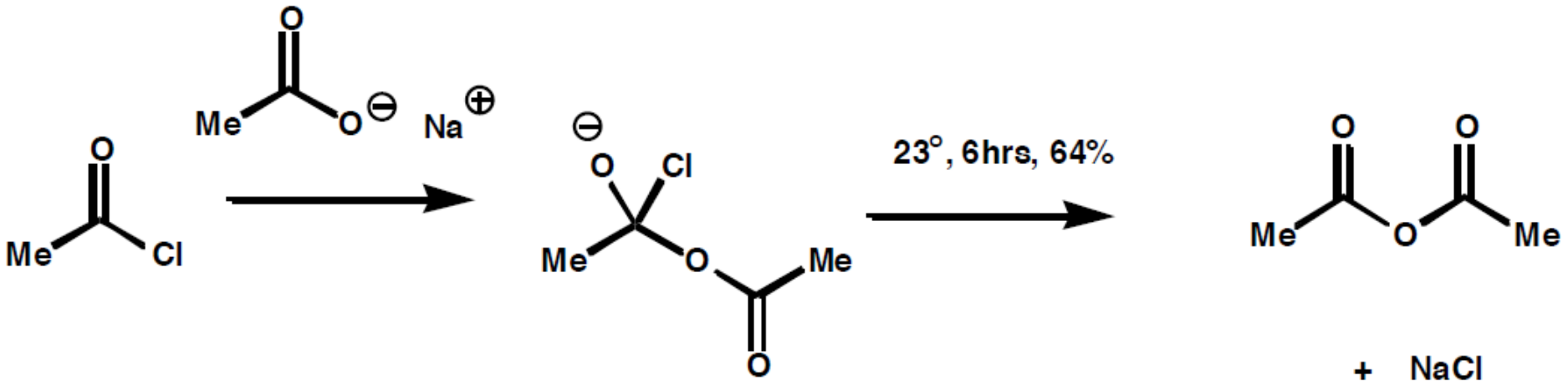
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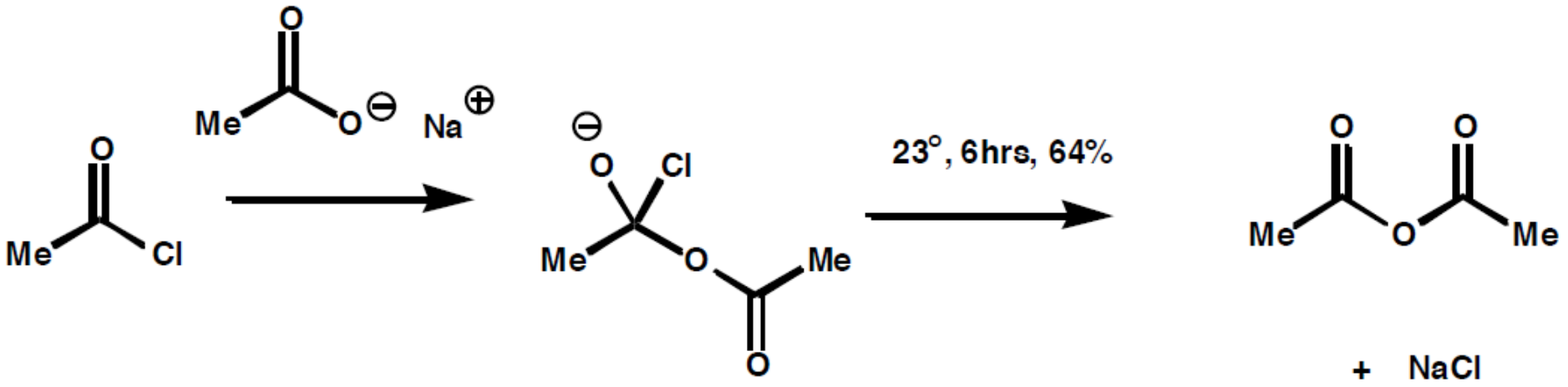
Example



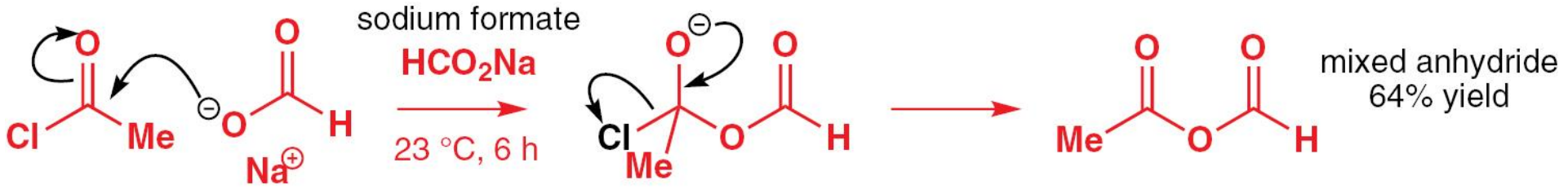
Making acid anhydrides from acyl chlorides



Making acid anhydrides from acyl chlorides

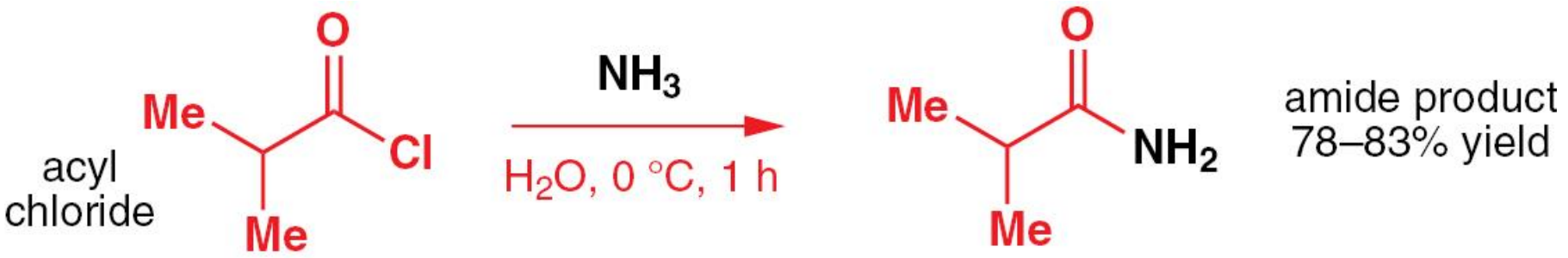


Example on synthesis of mixed anhydride

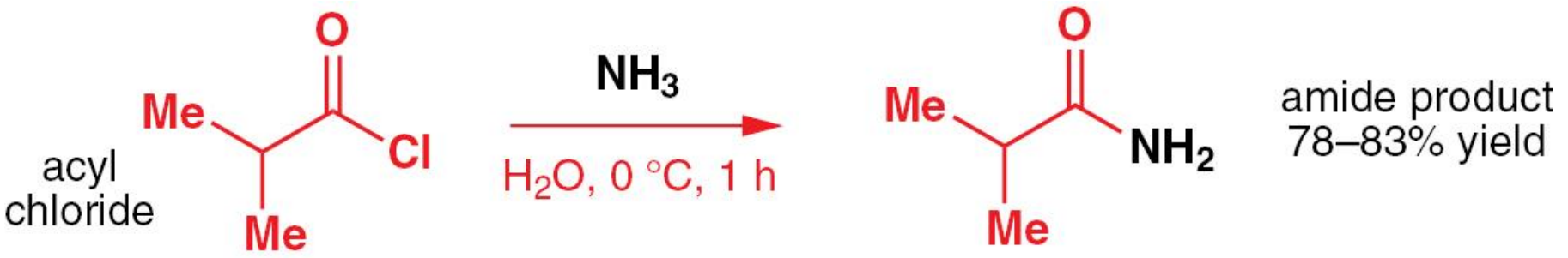


sodium methanoate

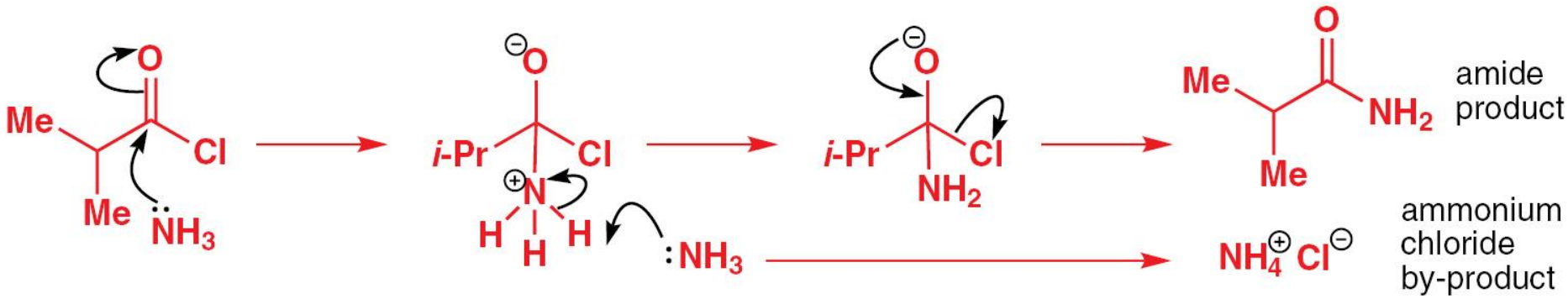
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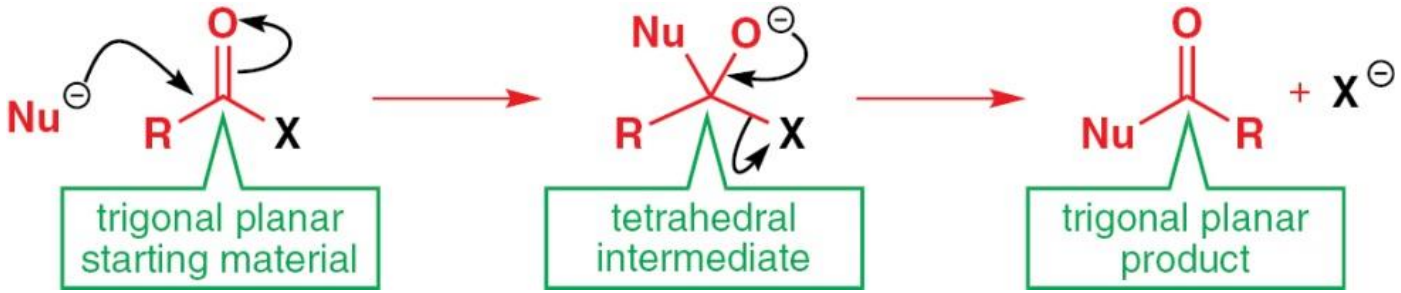
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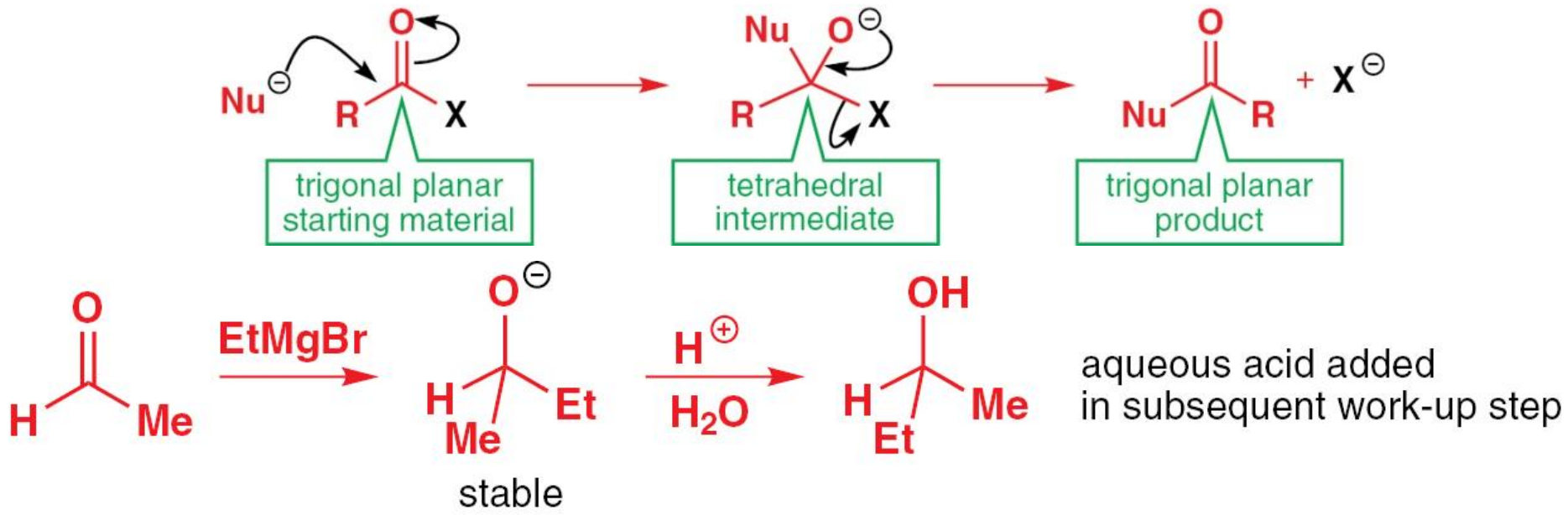
Mechanism for the reaction



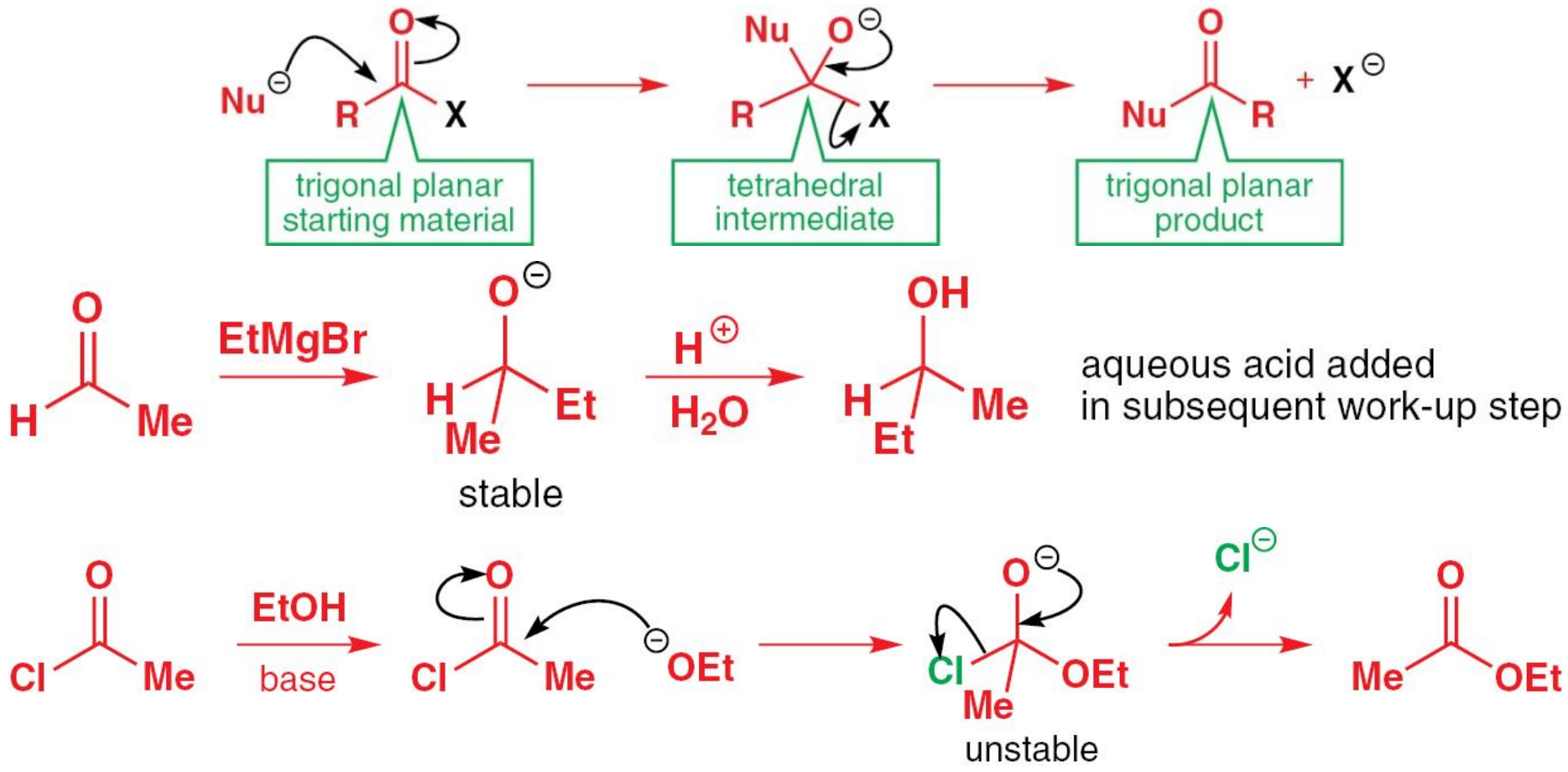
Tetrahedral intermediates



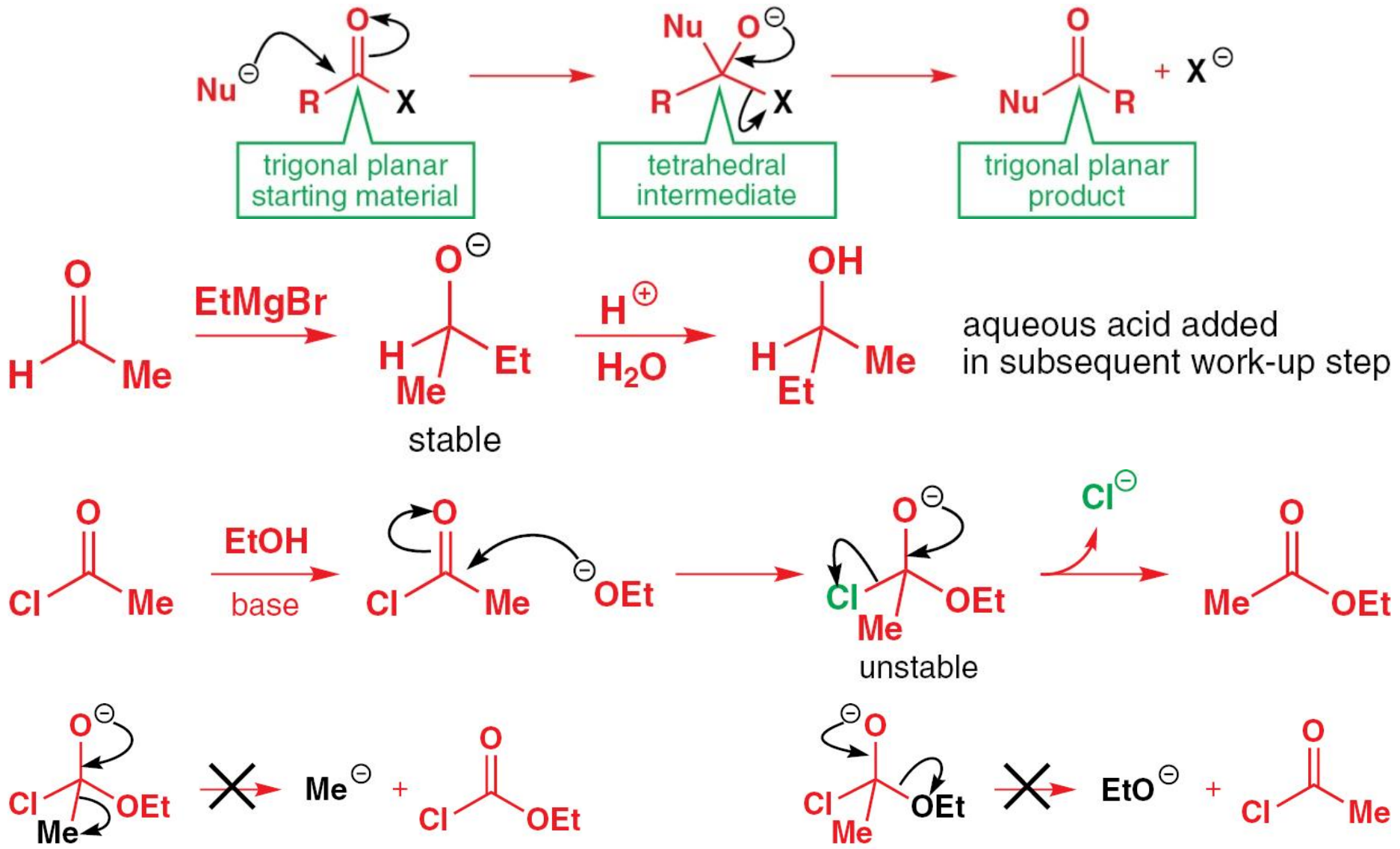
Tetrahedral intermediates



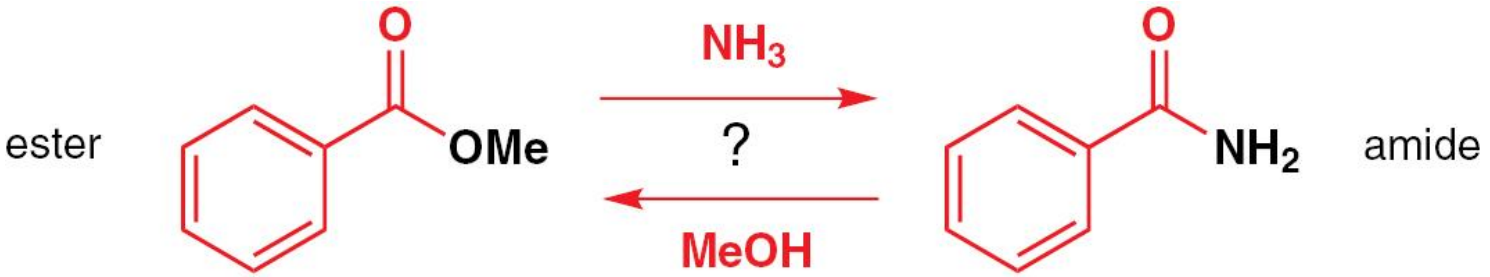
Tetrahedral intermediates



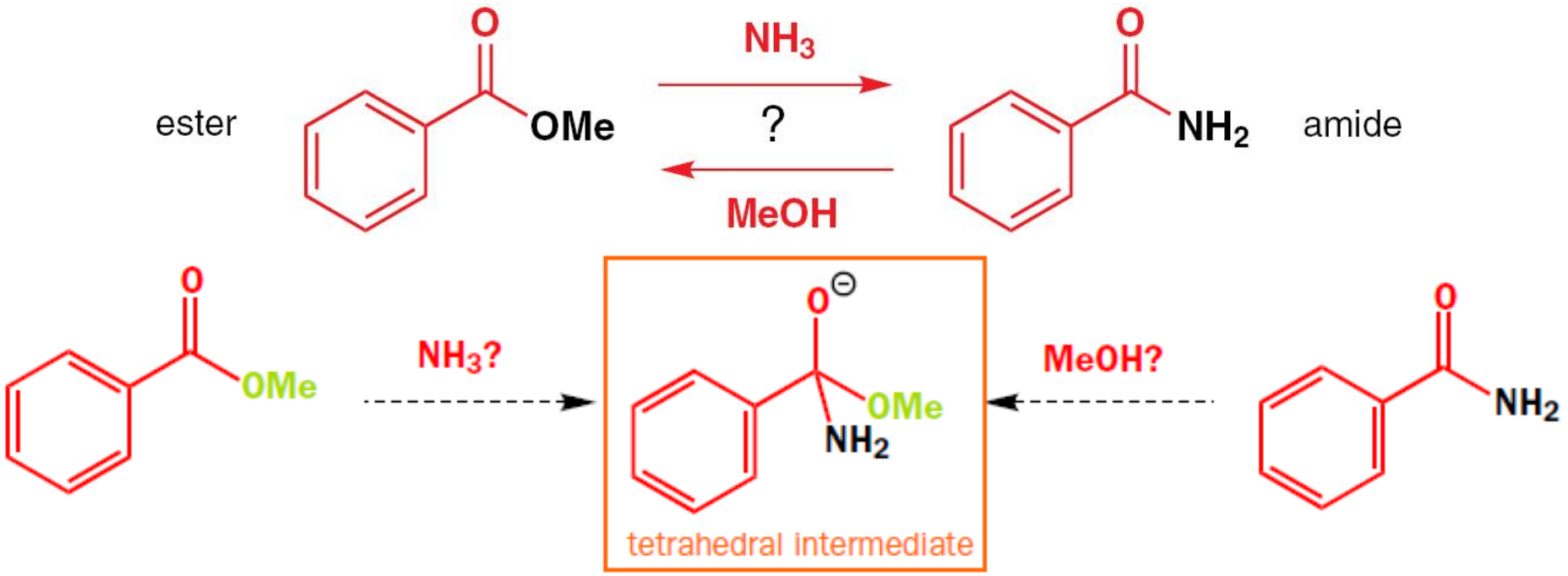
Tetrahedral intermediates



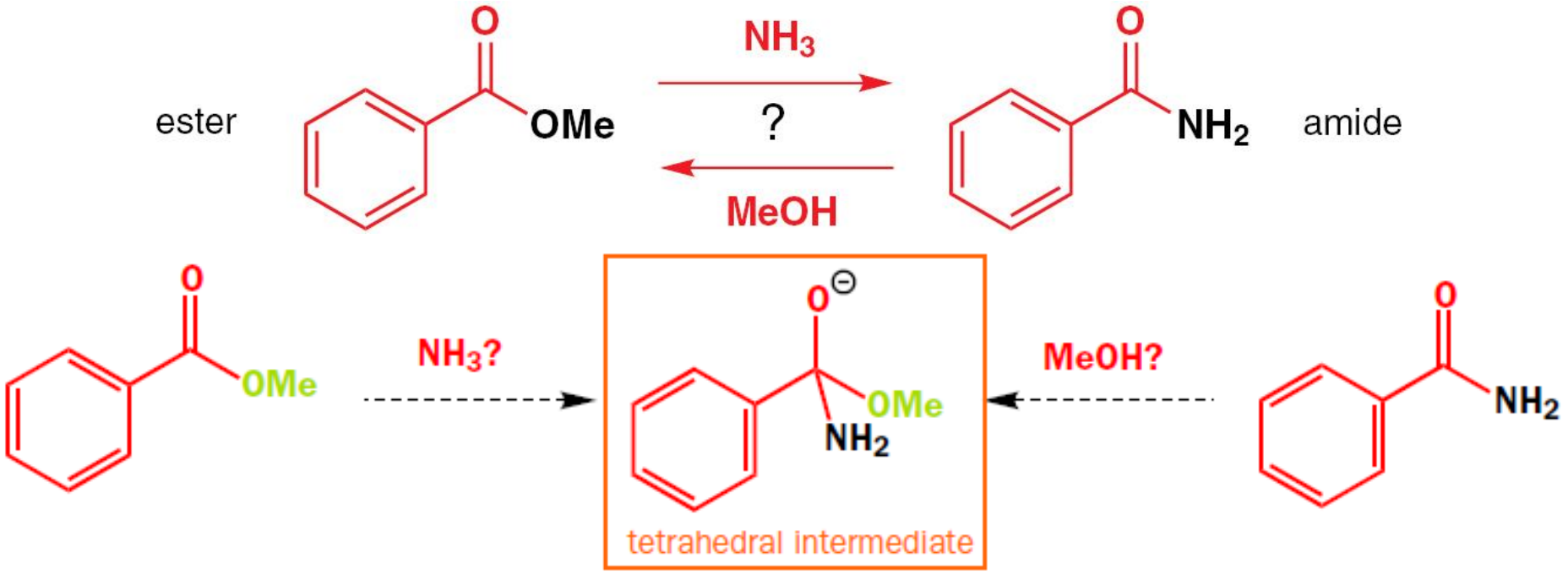
Leaving group ability: Can we make esters from amides?



Leaving group ability: Can we make esters from amides?

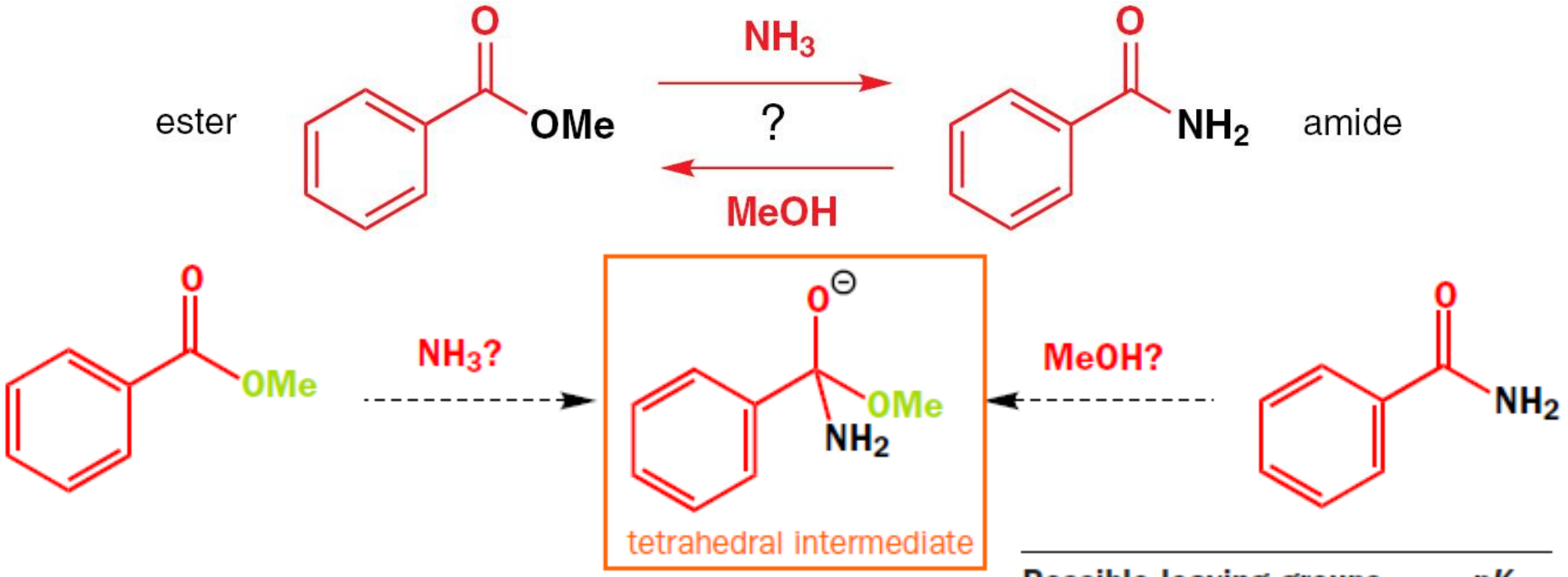


Leaving group ability: Can we make esters from amides?



A good leaving group can be defined as "the conjugate base of a strong acid"

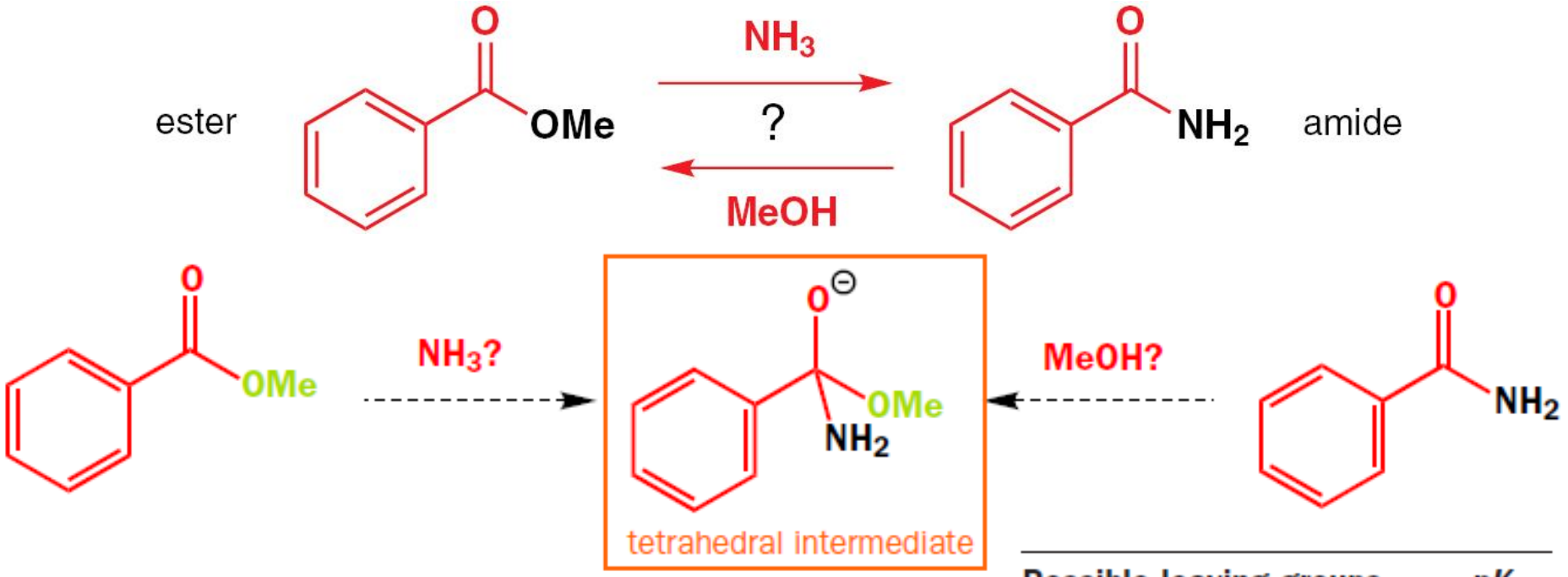
Leaving group ability: Can we make esters from amides?



Possible leaving groups	pK_{aH}
Ph^-	45
NH_2^-	35
MeO^-	16

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Leaving group ability: Can we make esters from amides?



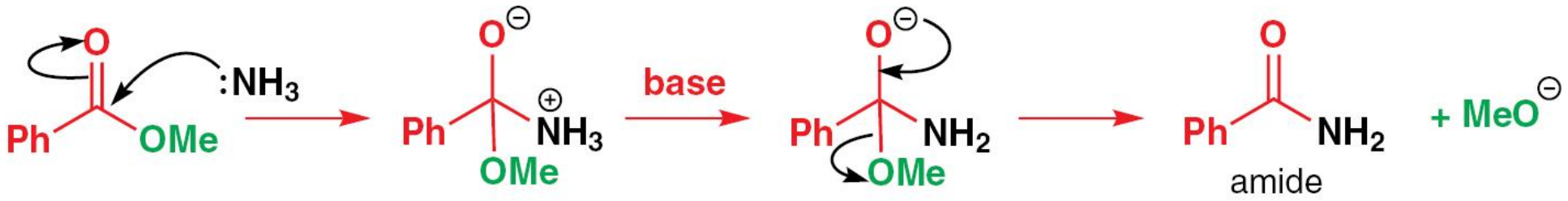
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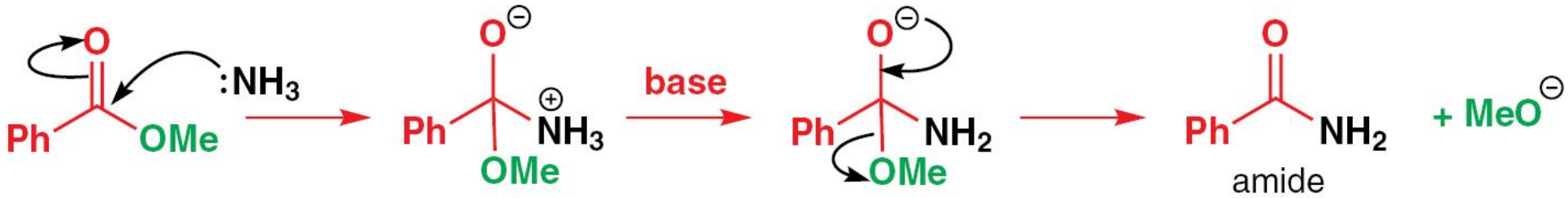
Leaving group ability

The lower the pK_{aH}, the better the leaving group in carbonyl substitution reactions.

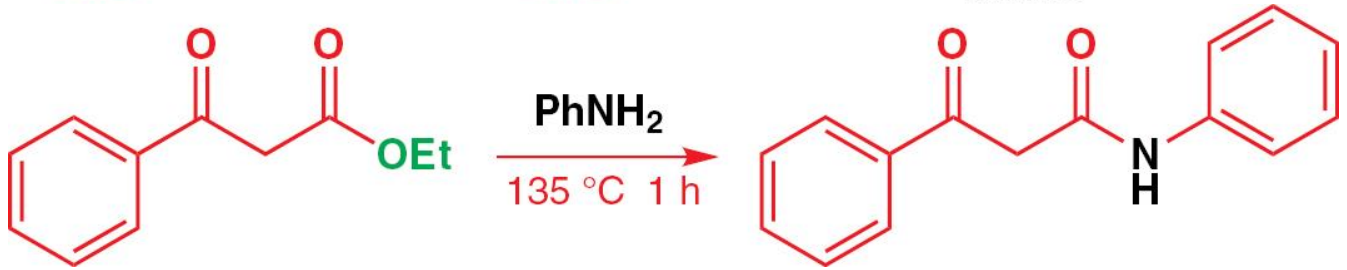
Making amides from esters



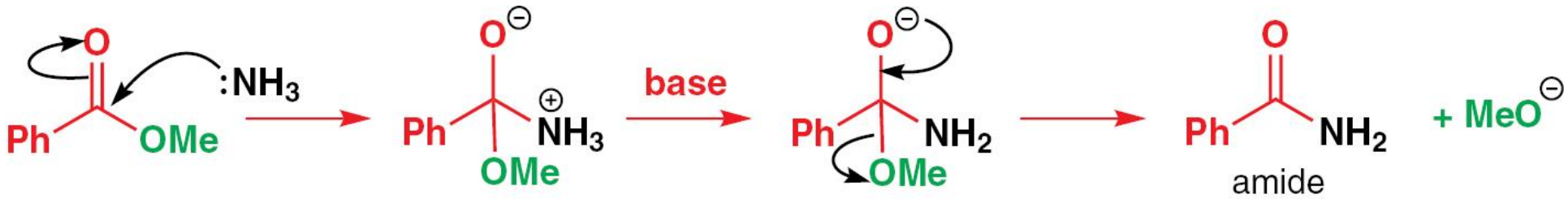
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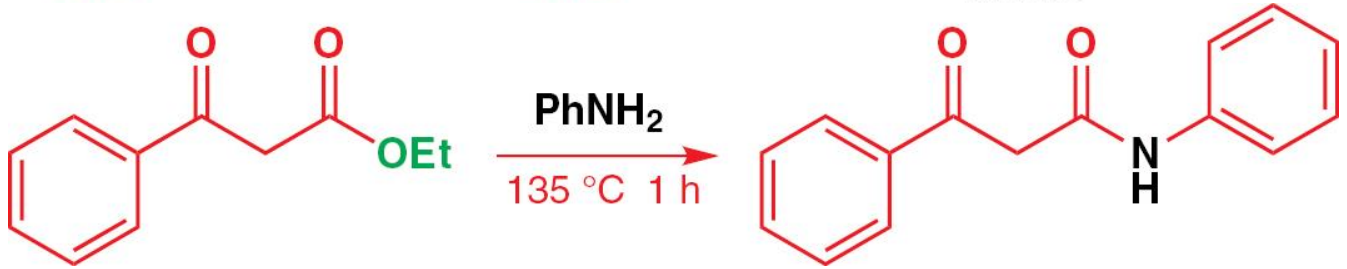
Example



Making amides from esters

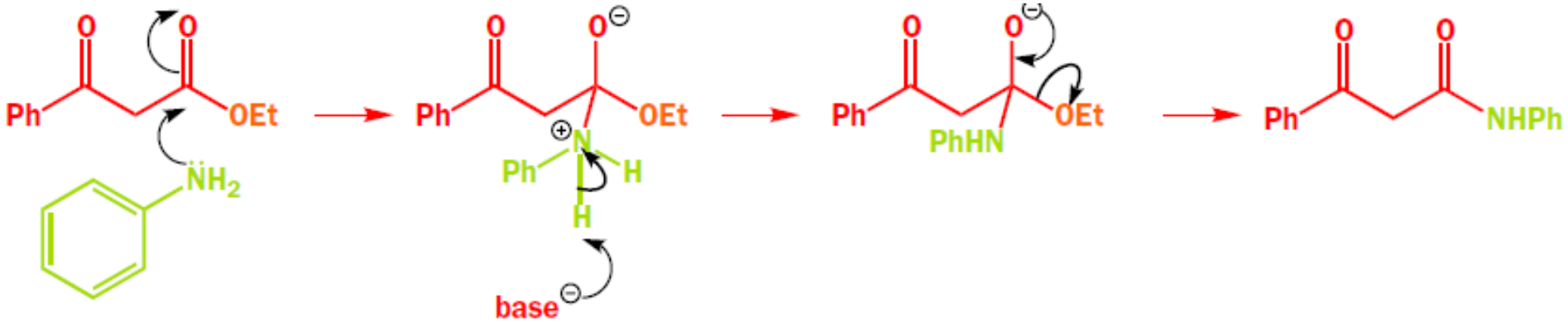


Example

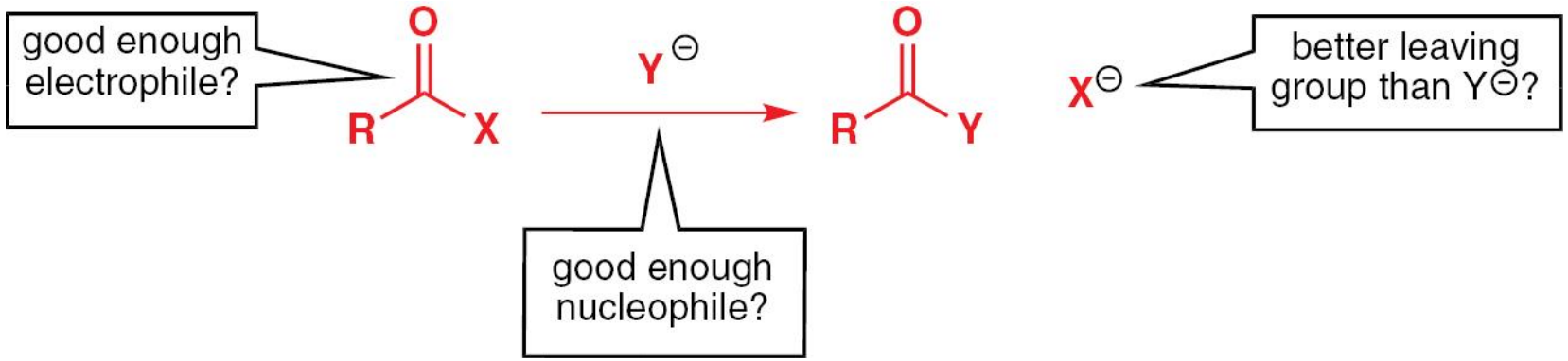


phenylamine

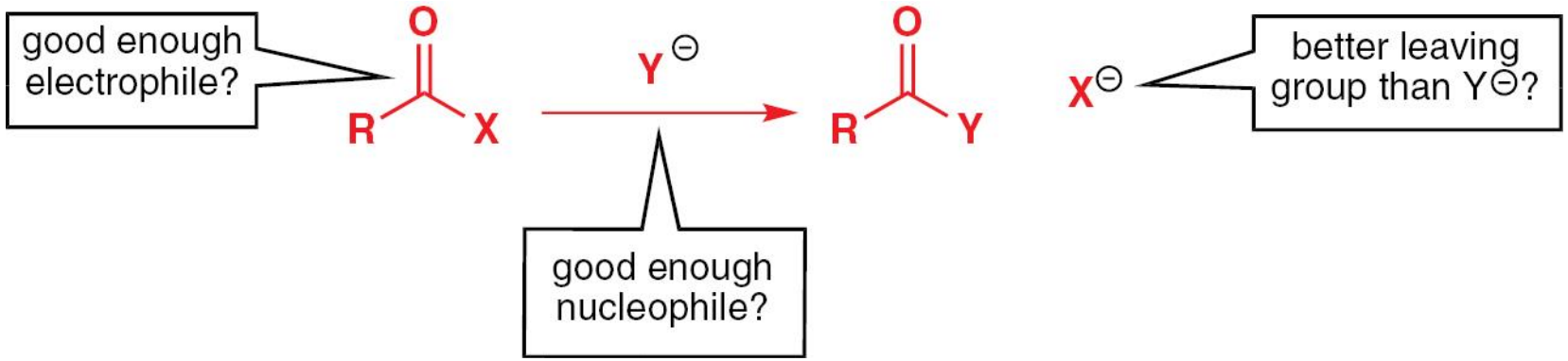
Mechanism for the reaction



Important factors except leaving group ability

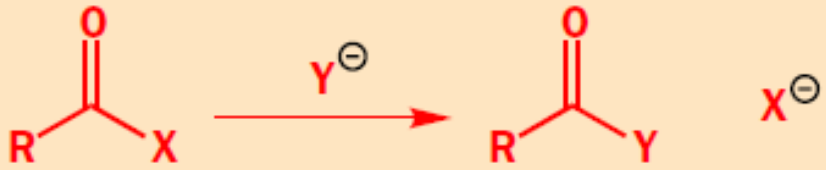


Important factors except leaving group ability



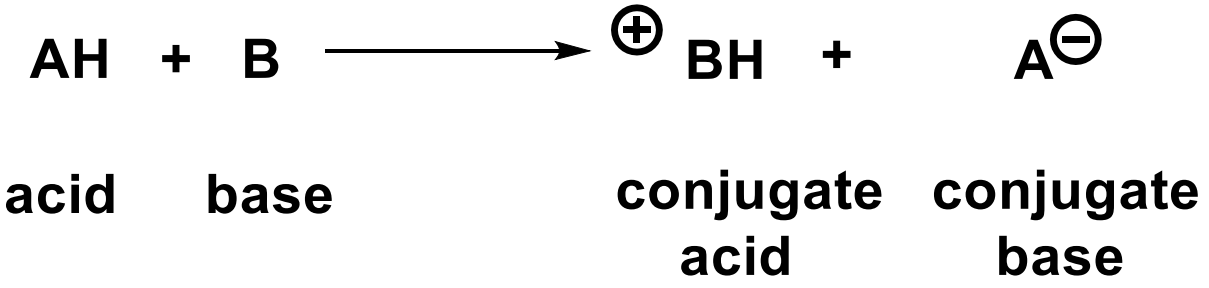
Conditions for reaction

If this reaction is to go



1. X must be a better leaving group than Y (otherwise the reverse reaction would take place)
2. Y must be a strong enough nucleophile to attack RCOX
3. RCOX must be a good enough electrophile to react with Y^-

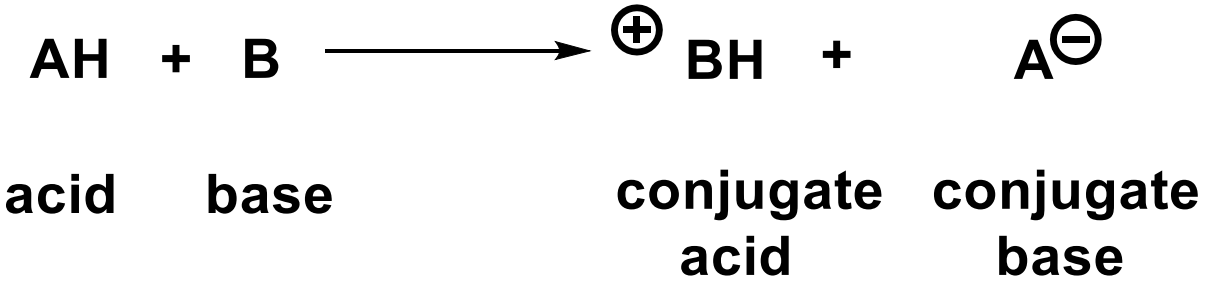
Guide to Nucleophilicity



Good nucleophiles are poor leaving groups

Base	p*K*aH
NH_3	9
NH_2^-	35
RO^-	16
Cl^-	-7
$RCOO^-$	5
Me^-	50
ROH	-5

Guide to Nucleophilicity



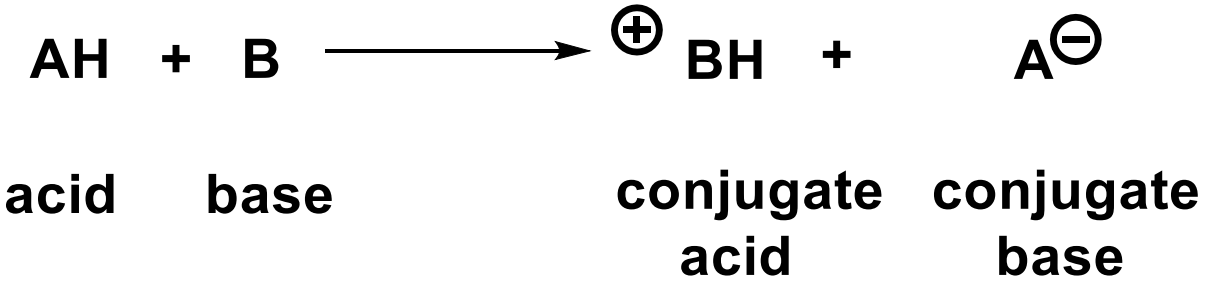
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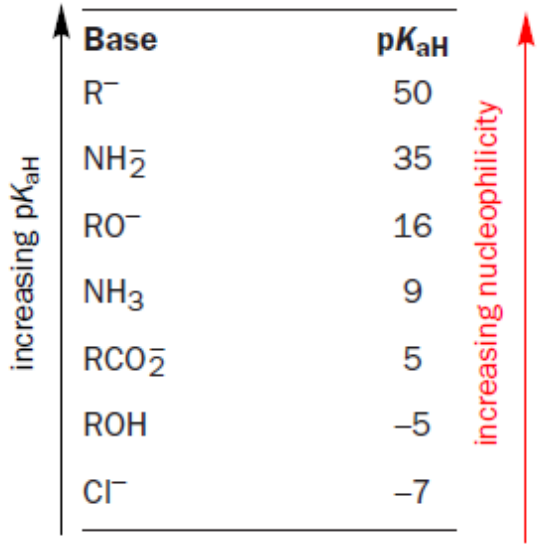
Guide to nucleophilicity

In general, the higher the pK_{aH}, the better the nucleophile.

Guide to Nucleophilicity



Good nucleophiles are poor leaving groups



Base	pK_{aH}
R^-	50
NH_2^-	35
RO^-	16
NH_3	9
RCO_2^-	5
ROH	–5
Cl^-	–7

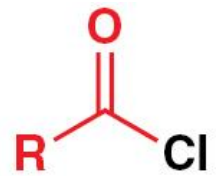
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Guide to nucleophilicity

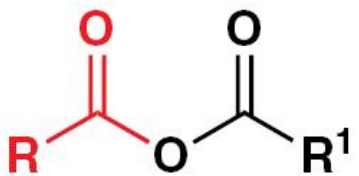
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Electrophiles

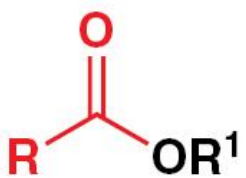
acid (acyl) chloride



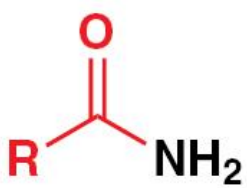
acid anhydride



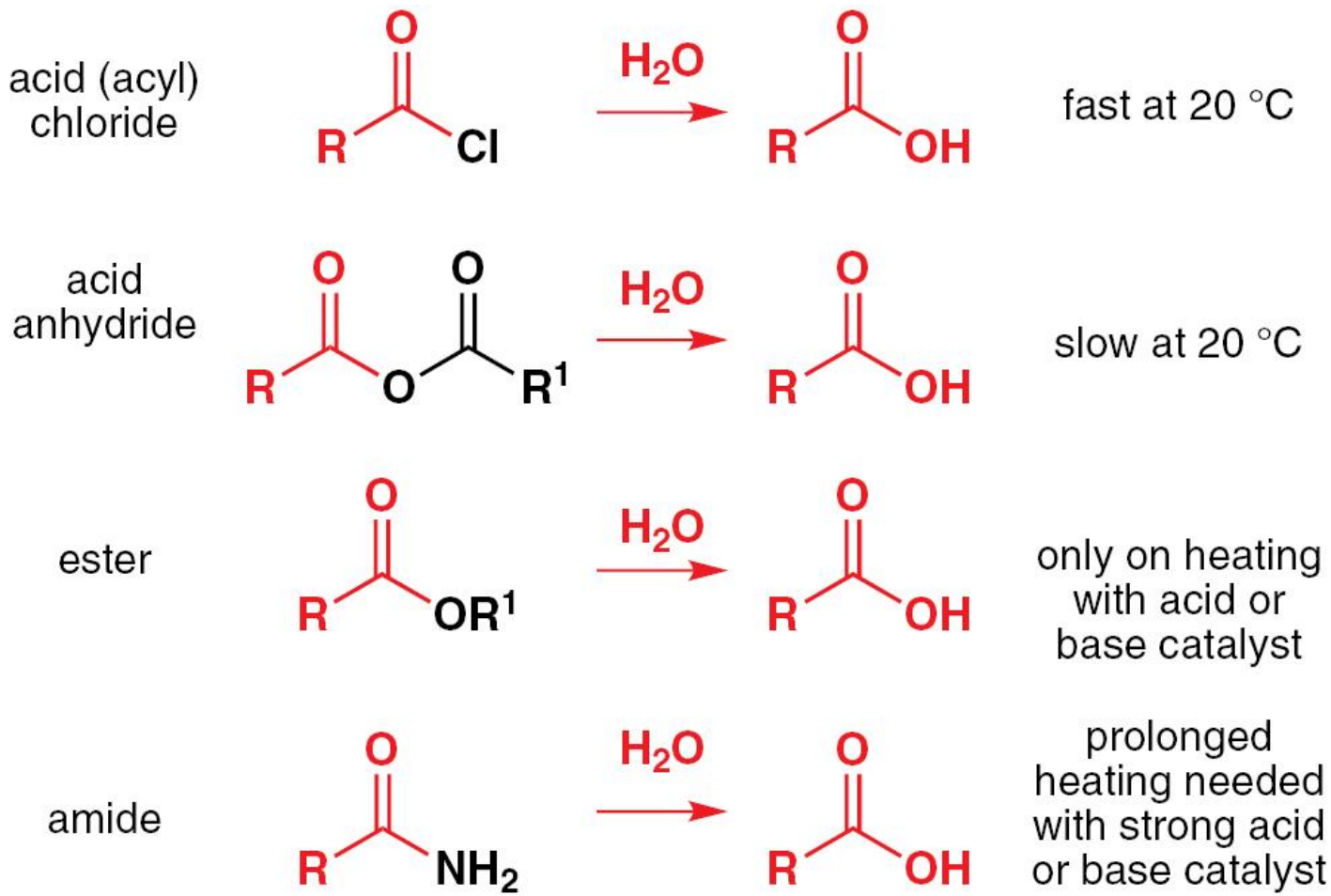
ester



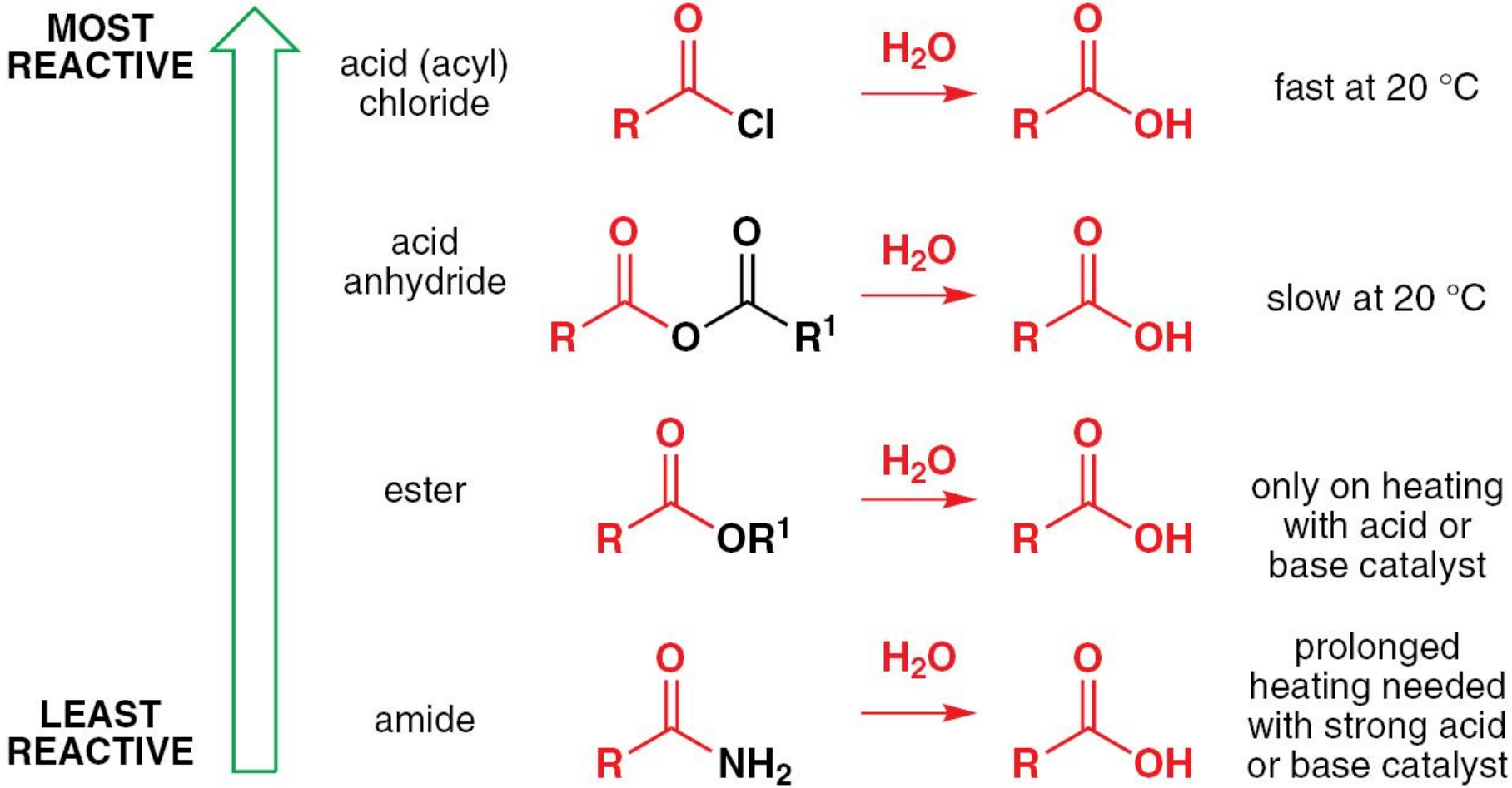
amide



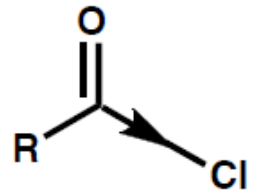
Electrophiles



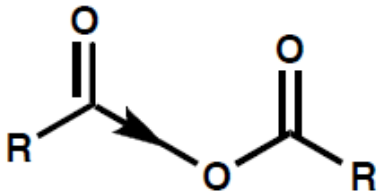
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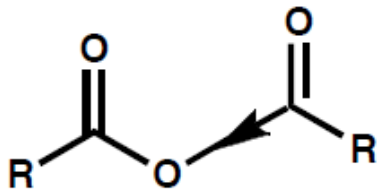


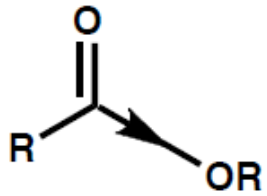
Inductive Effects

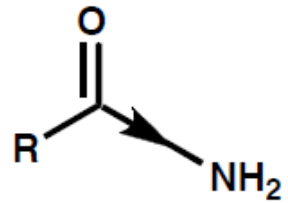


STRONGEST
INDUCTIVE EFFECT



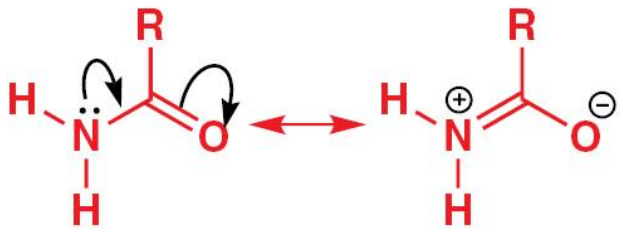


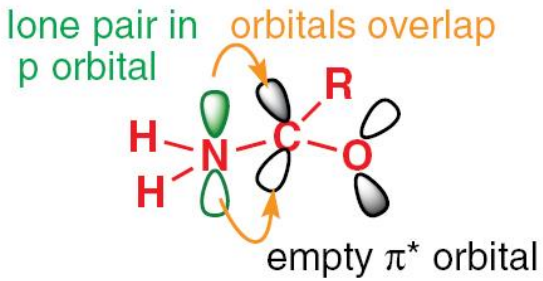


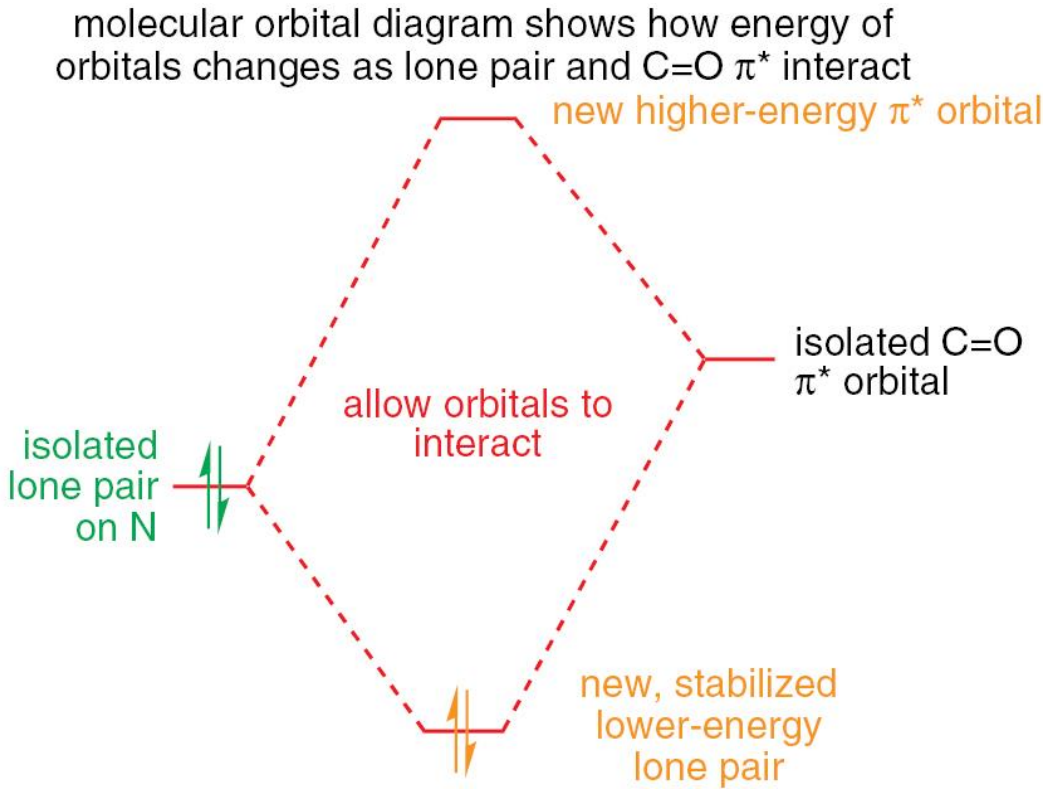


WEAKEST
INDUCTIVE EFFECT

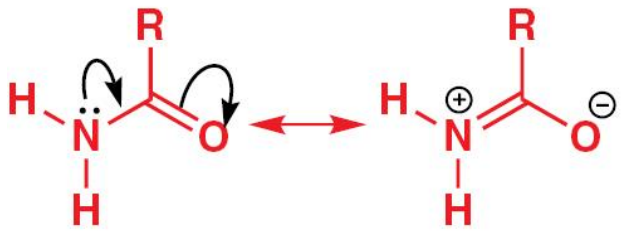
Conjugation Effects

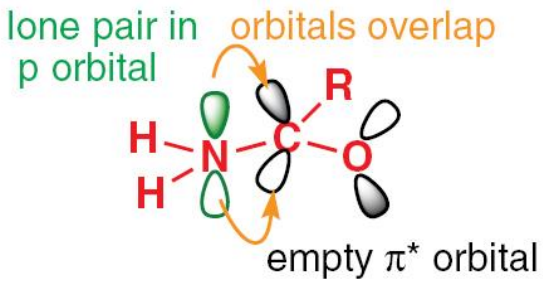




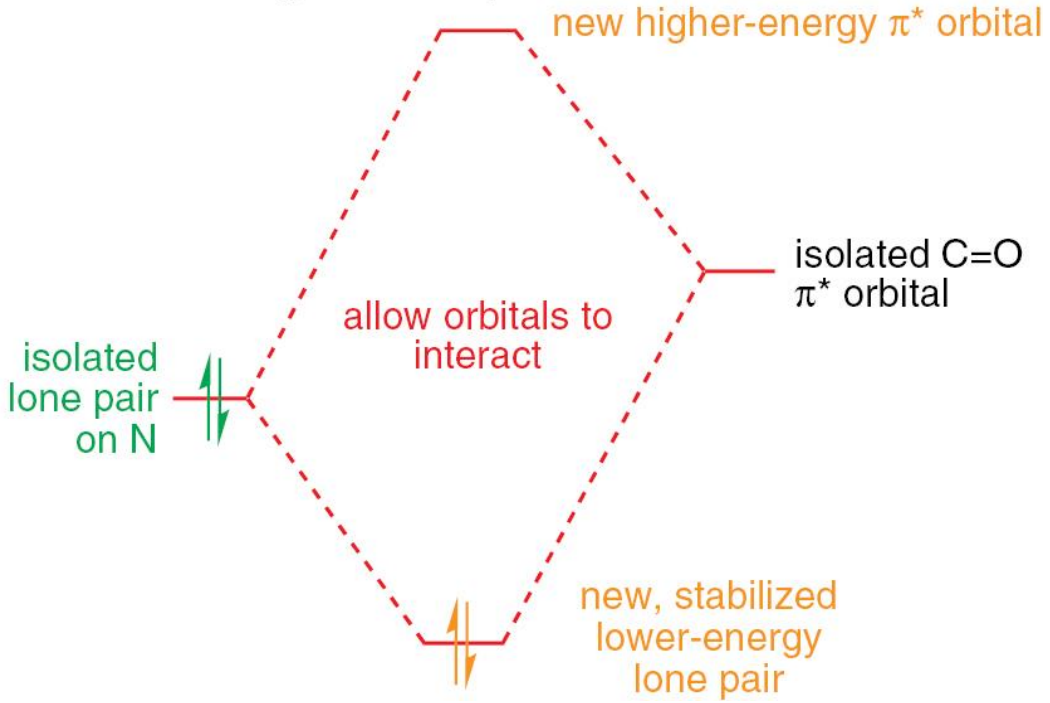


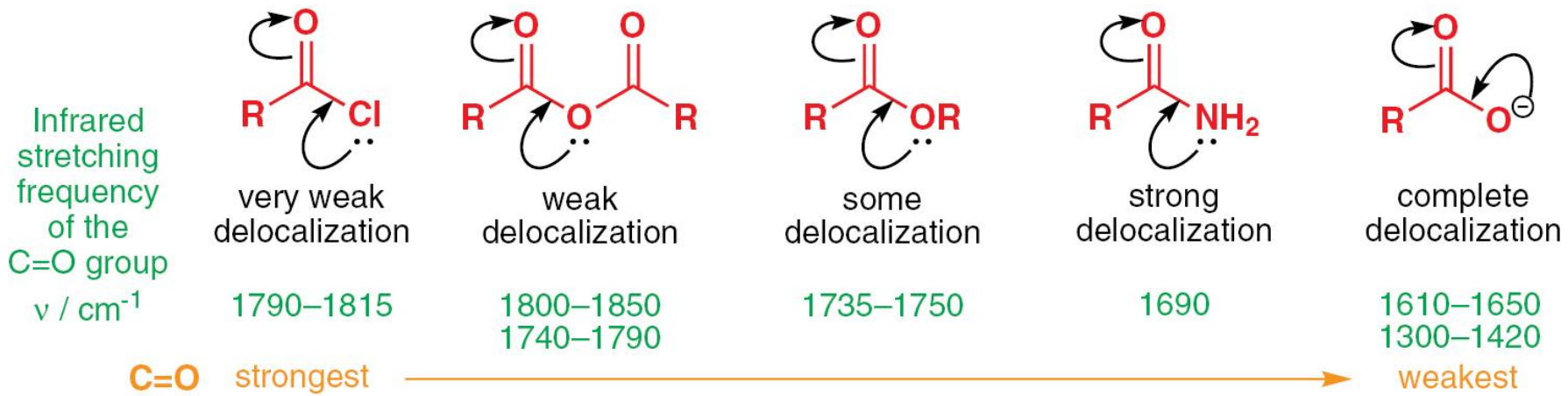
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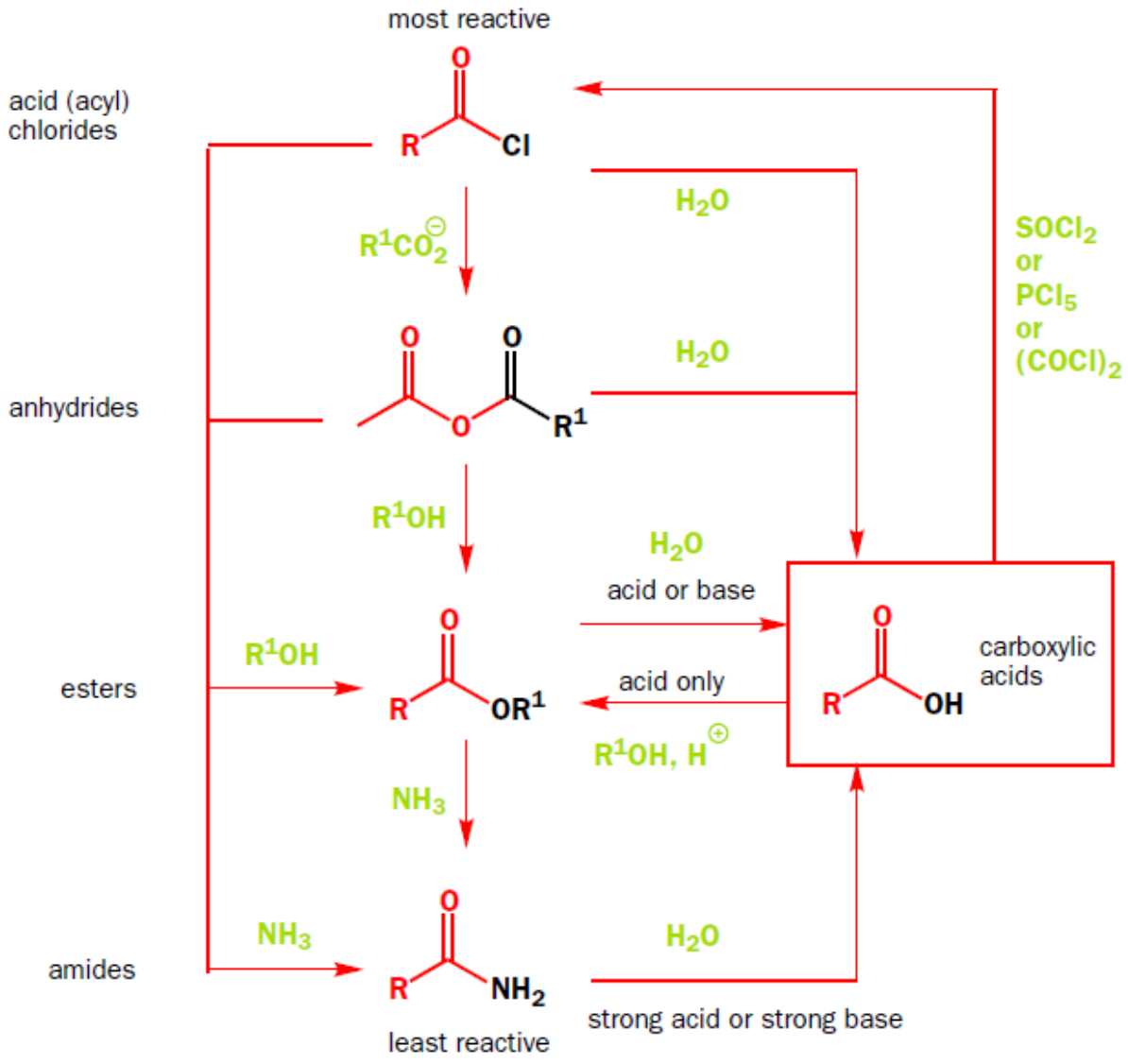


molecular orbital diagram shows how energy of orbitals changes as lone pair and C=O π* interact





Reactivity Chart



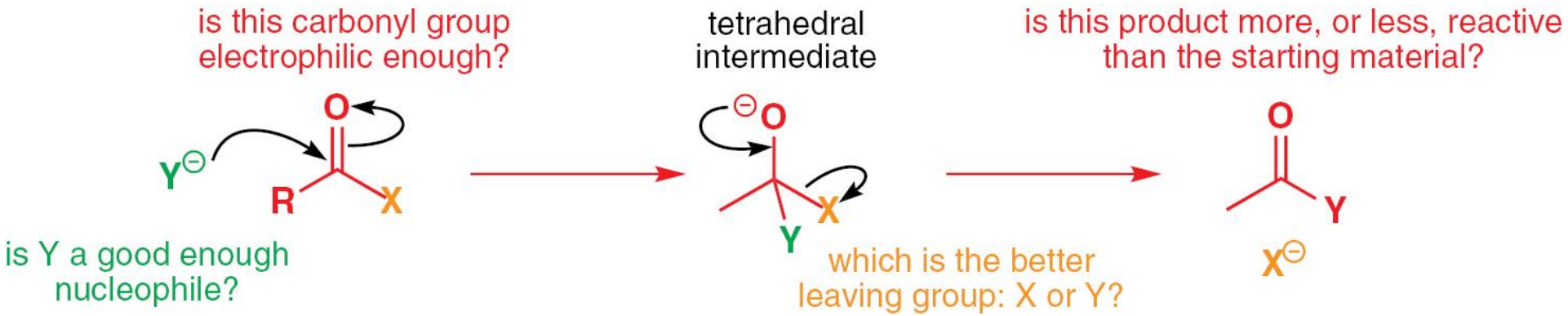
Summary

- Nucleophilic substitution at the carbonyl group takes place through tetrahedral intermediates
- Some leaving groups are better than others
- Some nucleophiles are better than others
- Some electrophiles are better than others

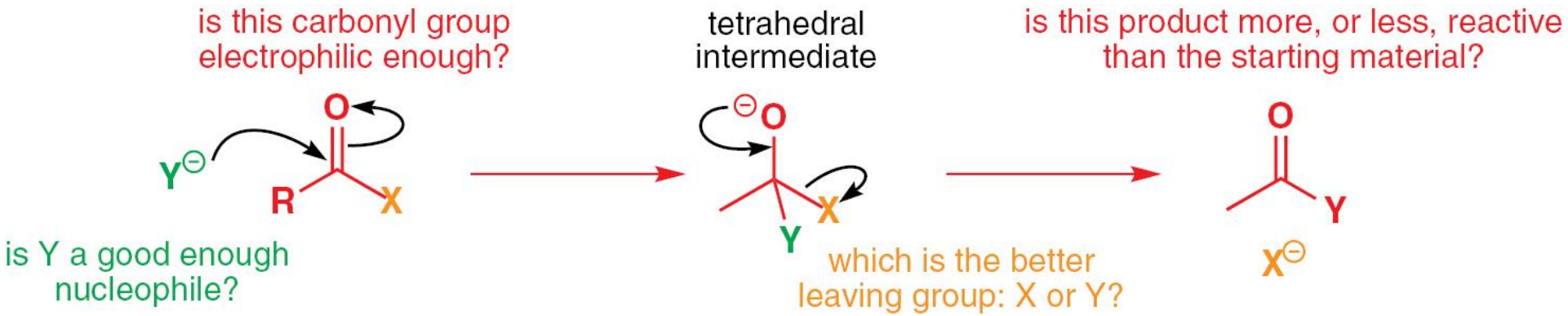
Nucleophilic *Substitution* at the Carbonyl Group

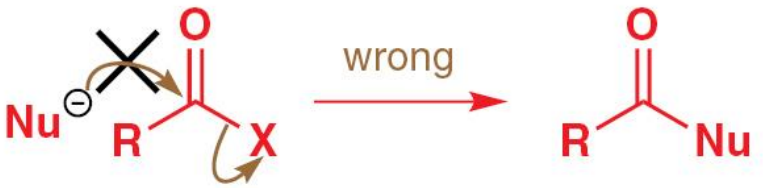
- Nucleophilic attack followed by loss of leaving group
- What makes a good nucleophile
- What makes a good leaving group
- There is always a tetrahedral intermediate
- **How to make acid derivatives**
- **Reactivity of acid derivatives**
- **How to make ketones from acids**
- **How to reduce acids to alcohols**

Nucleophilic substitution reactions

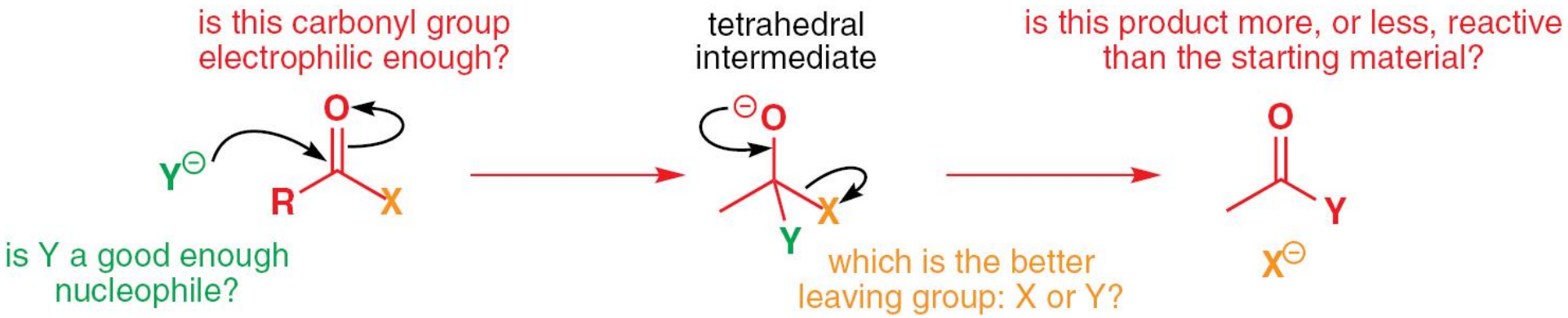


Nucleophilic substitution reactions

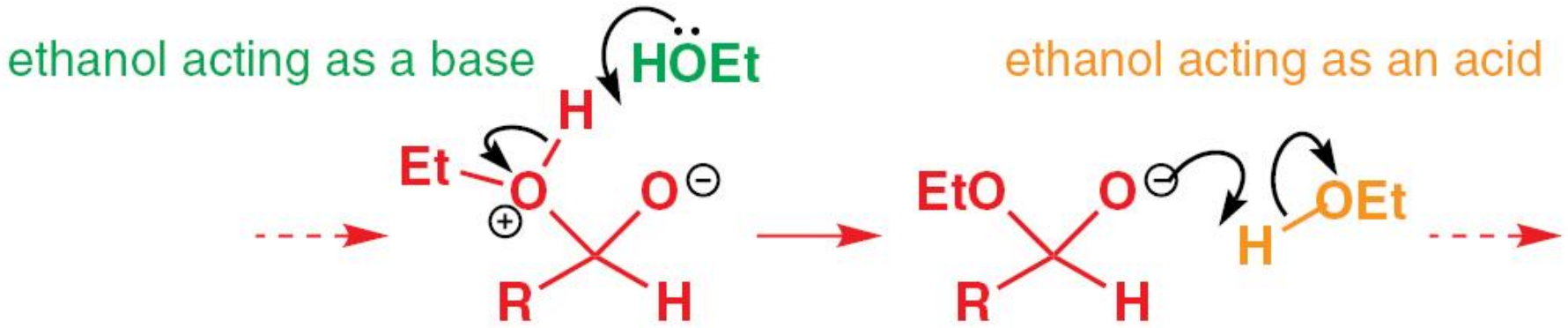




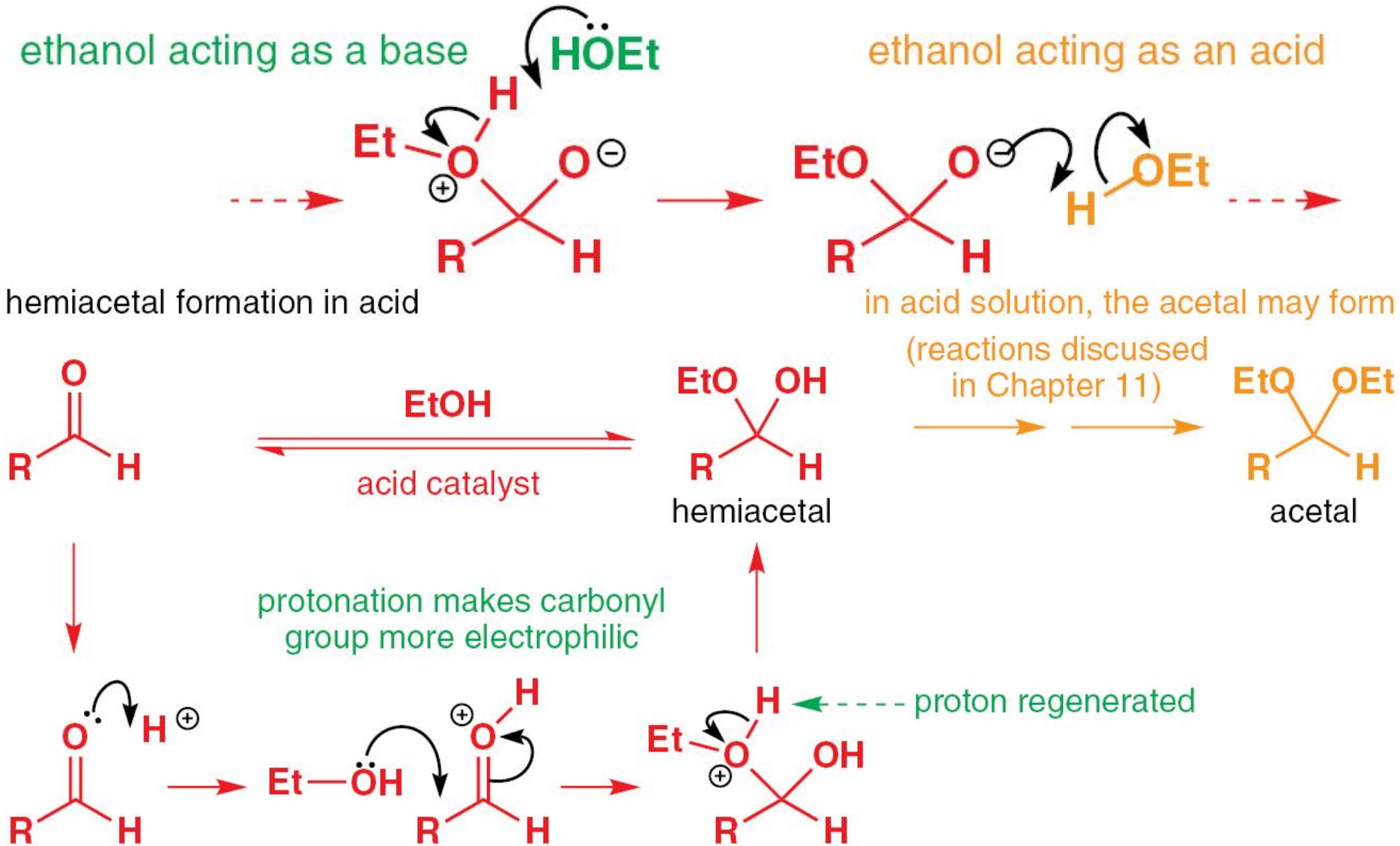
Nucleophilic substitution reactions



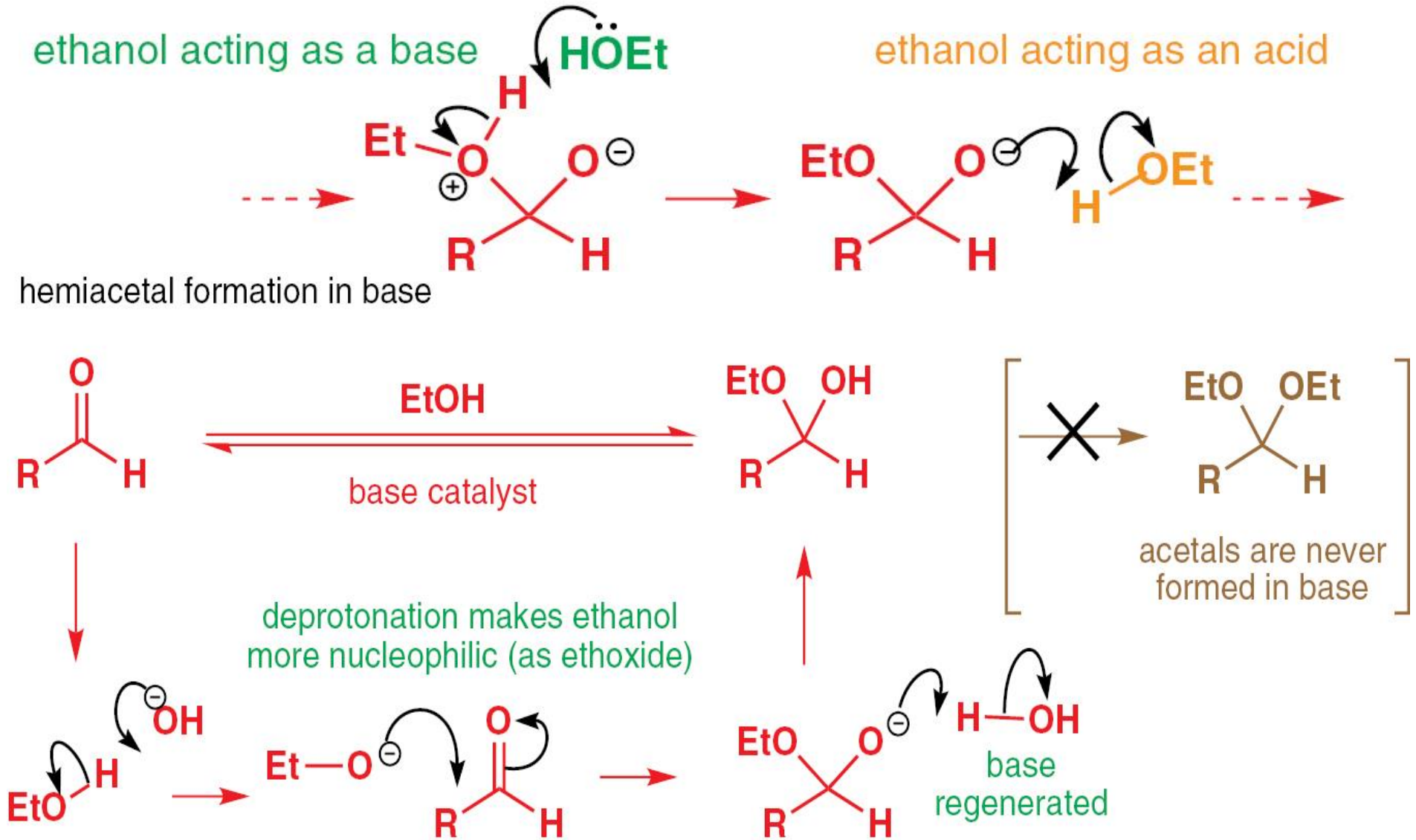
Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision



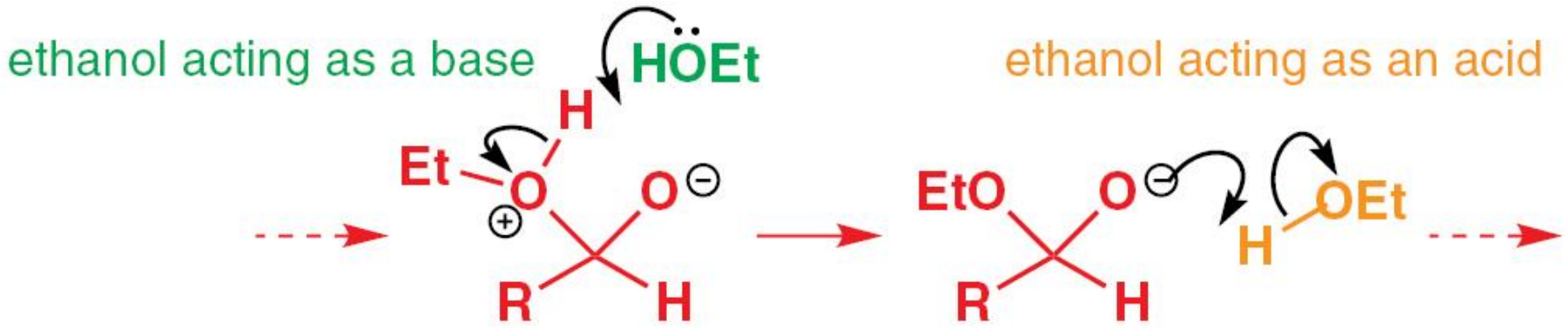
Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision



Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision

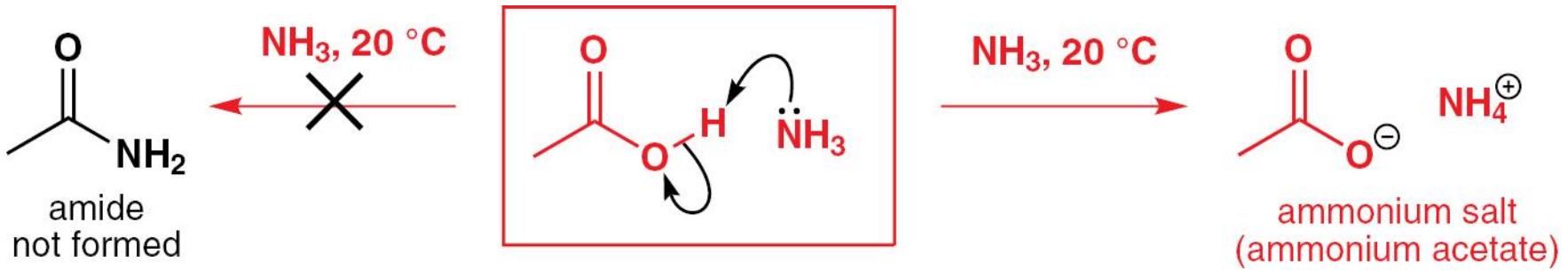


Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision

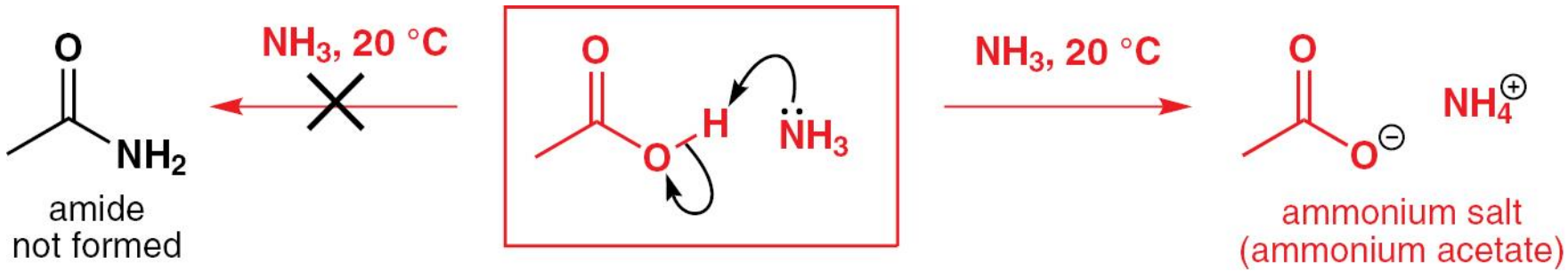


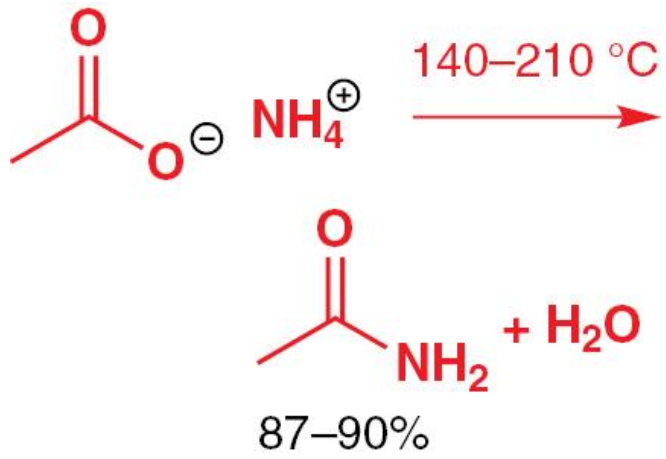
- For nucleophilic additions to carbonyl groups:
  - Acid catalysts work by making the carbonyl group more electrophilic
  - Base catalysts work by making the nucleophile more nucleophilic

Why carboxylic acids do not undergo substitution reactions under basic conditions?



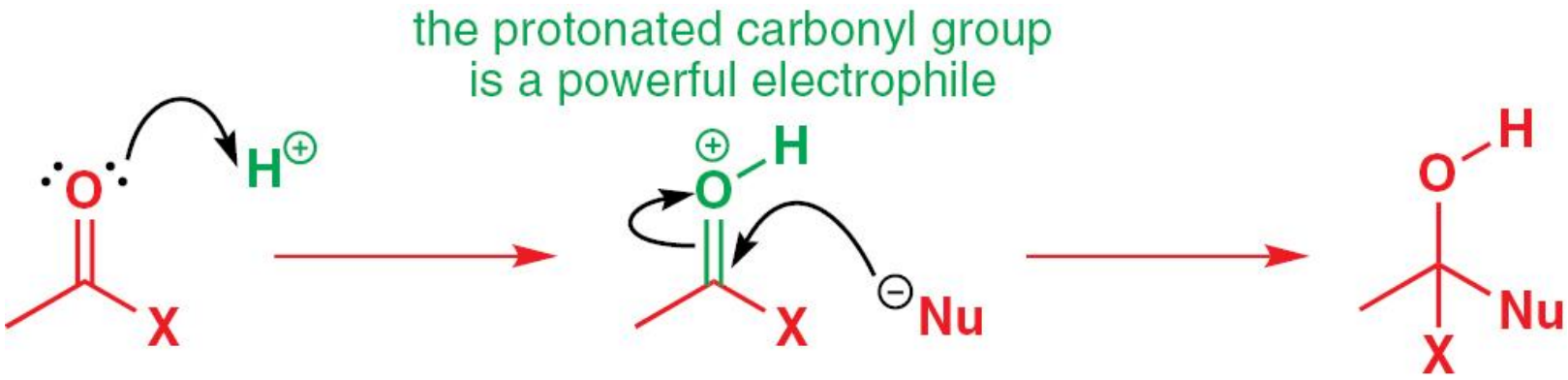
Why carboxylic acids do not undergo substitution reactions under basic conditions?





How acid catalysts increase the reactivity of the carbonyl group?

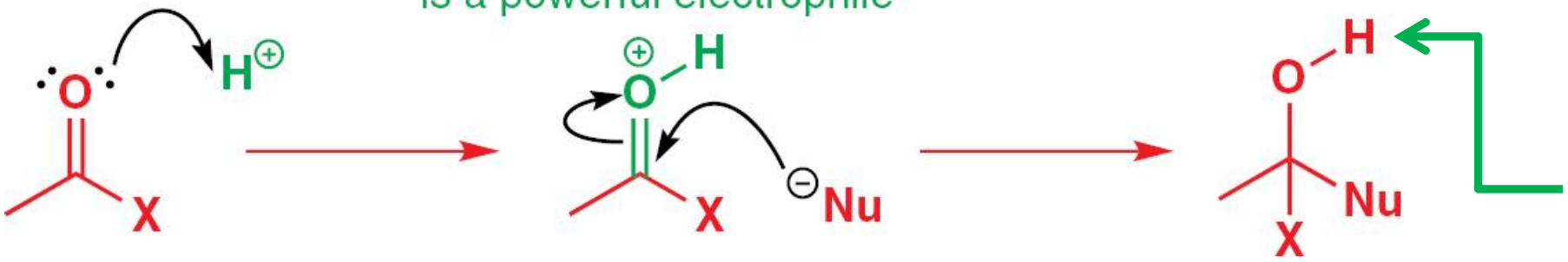
- **Acid catalysts catalyse substitution reactions of carboxylic acids**
  1 They increase the electrophilicity of the carbonyl group by protonation at *carbonyl* oxygen



How acid catalysts increase the reactivity of the carbonyl group?

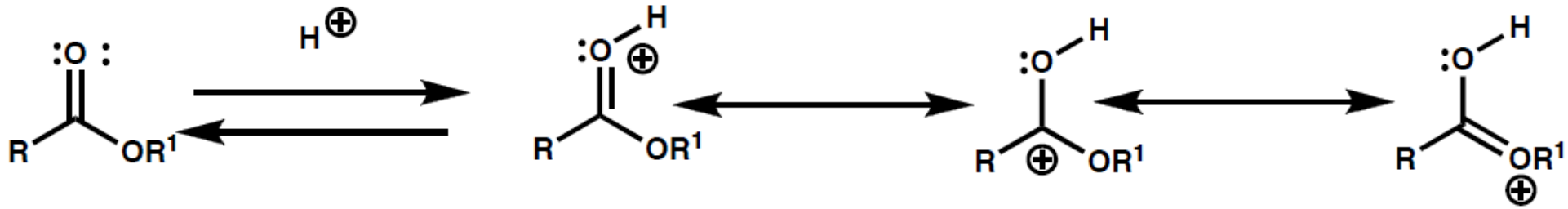
- **Acid catalysts catalyse substitution reactions of carboxylic acids**
  1. They increase the electrophilicity of the carbonyl group by protonation at *carbonyl* oxygen
  2. They lower the pK_{aH} of the leaving group by protonation there too

the protonated carbonyl group is a powerful electrophile

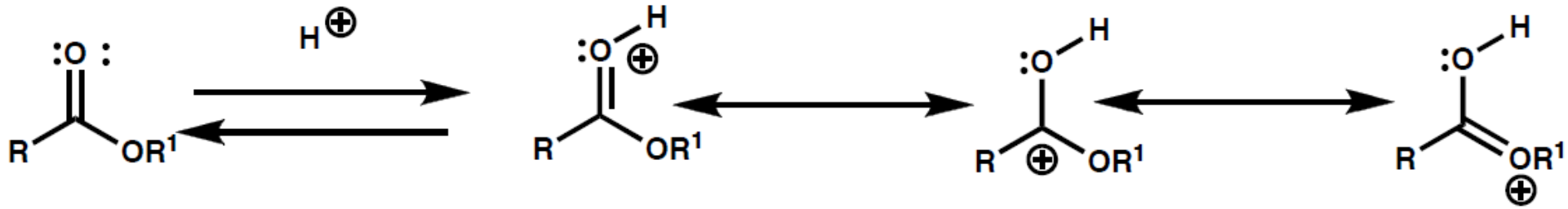


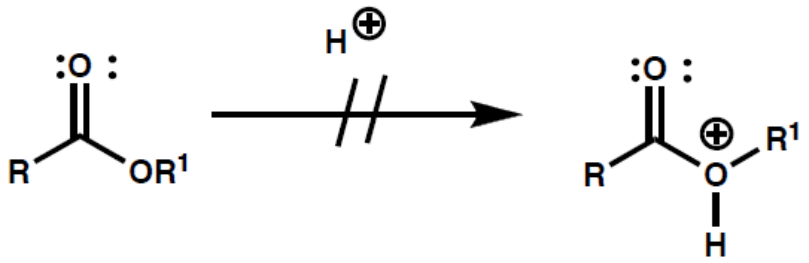
Protonation here will convert the poor leaving group OH^- to the good leaving group H_2O

Protonation occurs always on the carbonyl oxygen

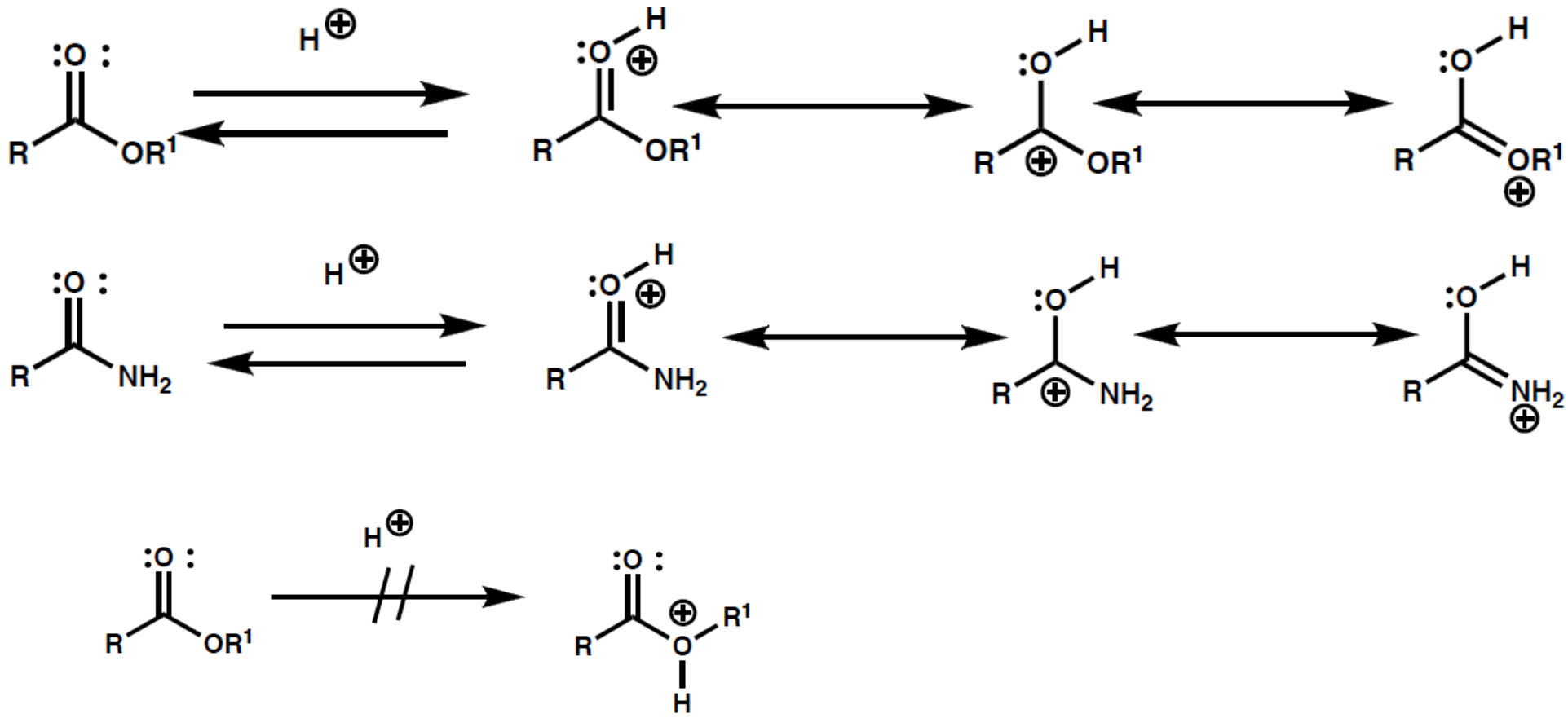


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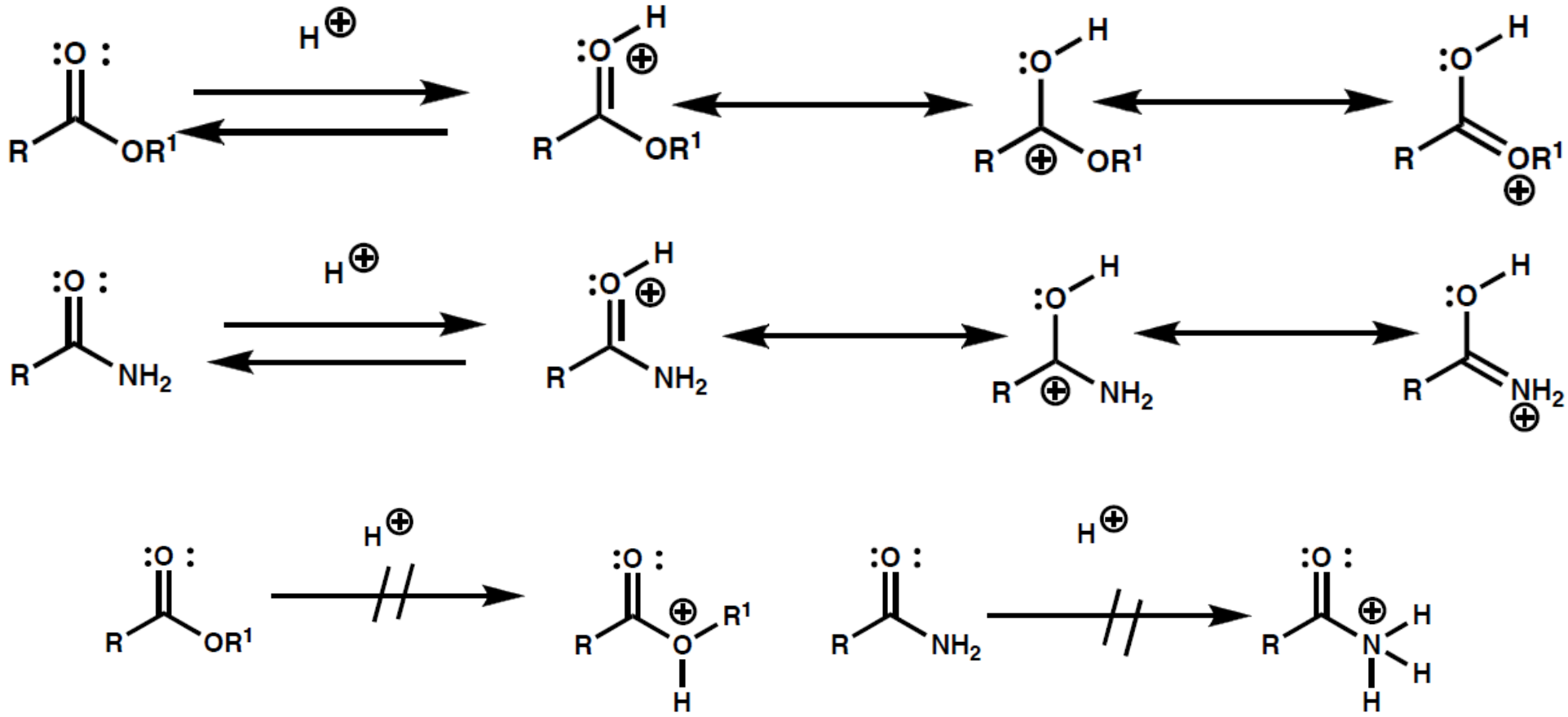


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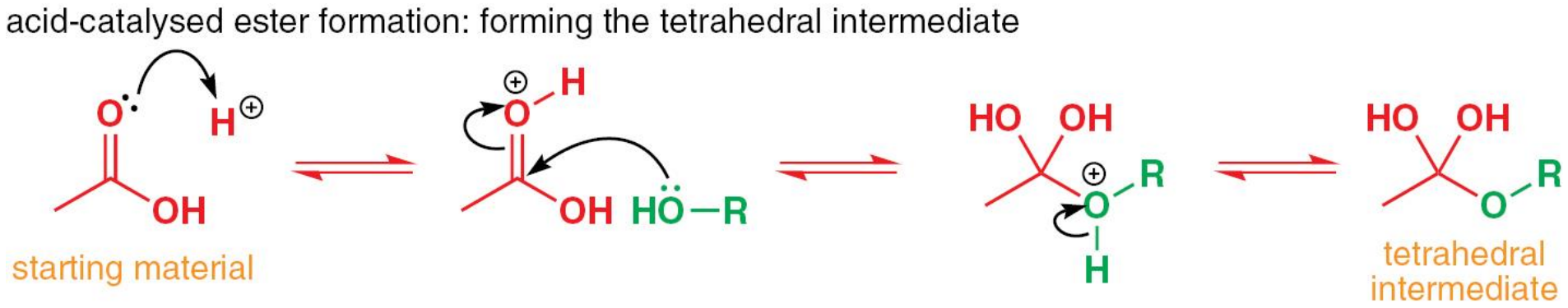
No delocalization possible

Protonation occurs always on the carbonyl oxygen

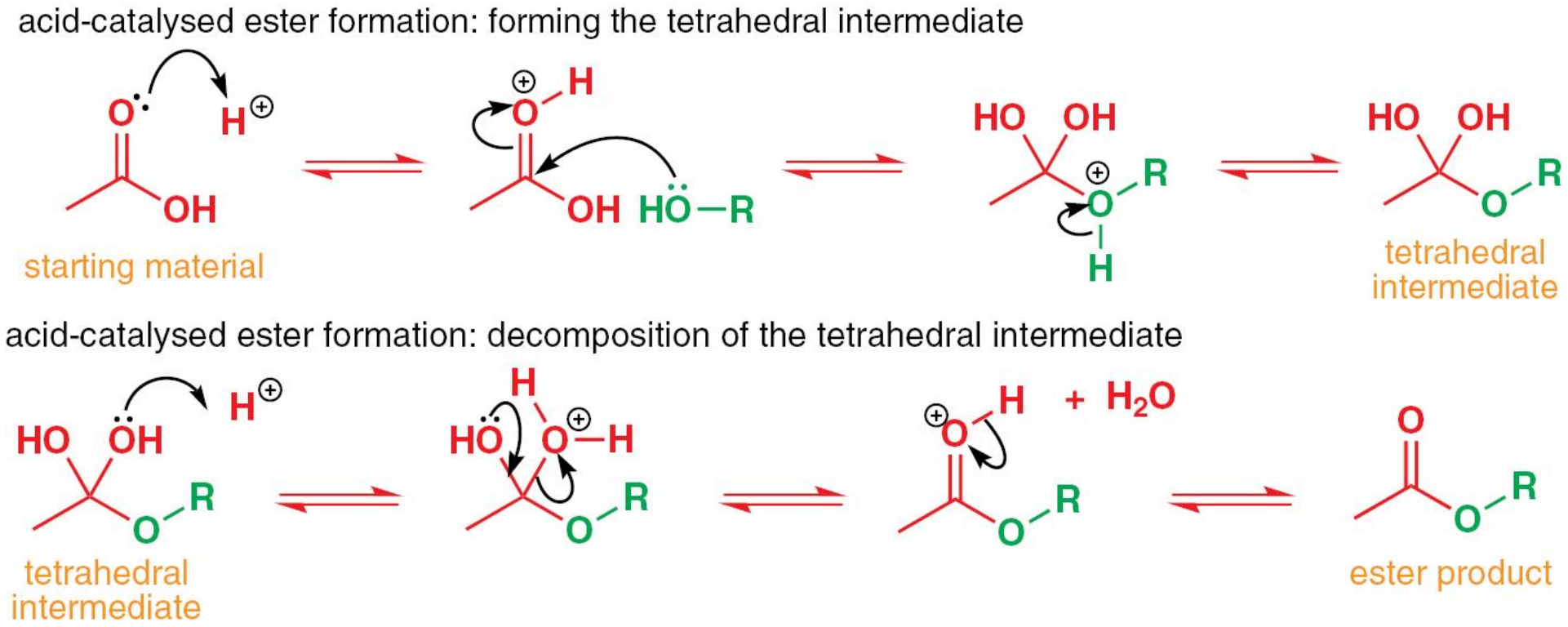


No delocalization possible

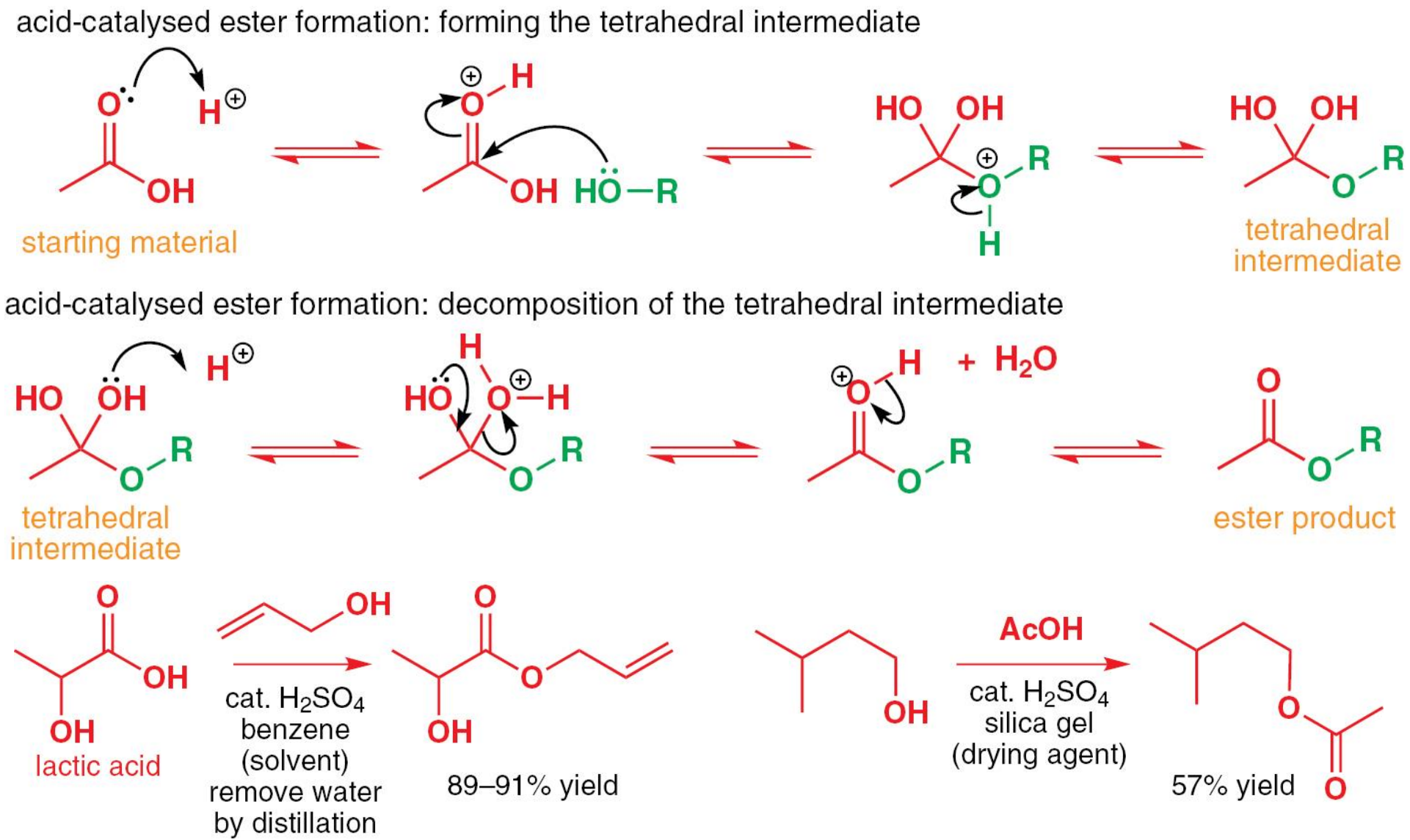
Acid-catalyzed ester formation



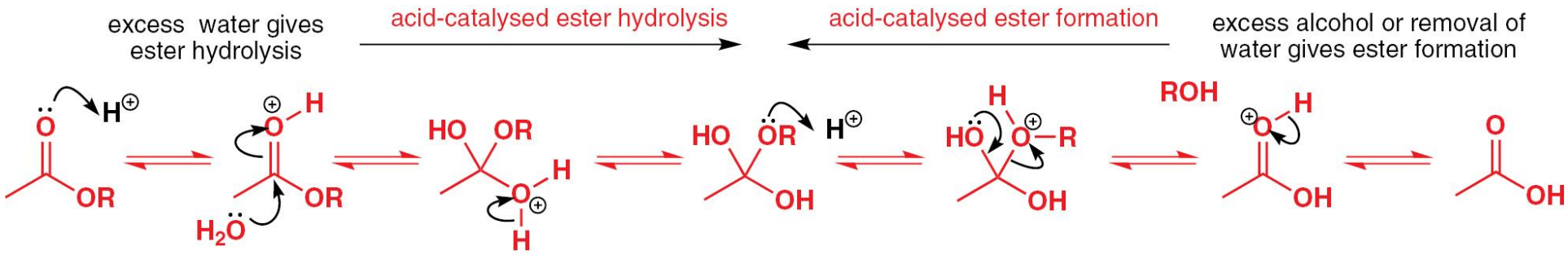
Acid-catalyzed ester formation



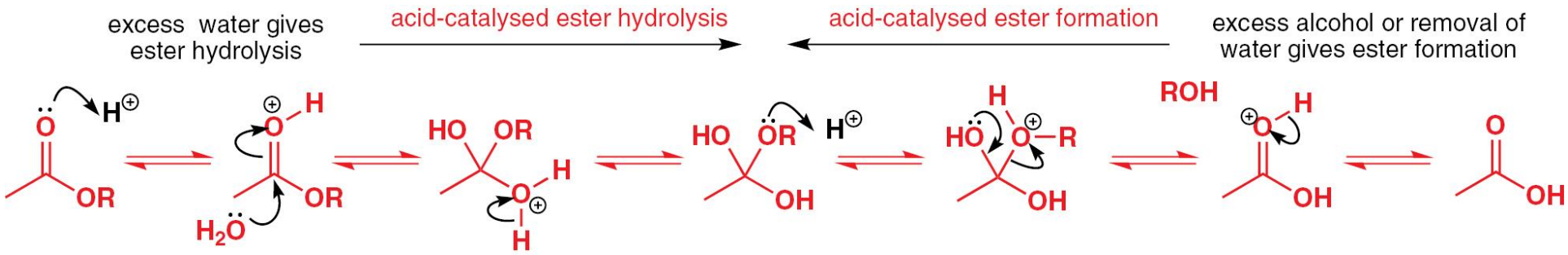
Acid-catalyzed ester formation



Acid-catalyzed ester hydrolysis and transesterification



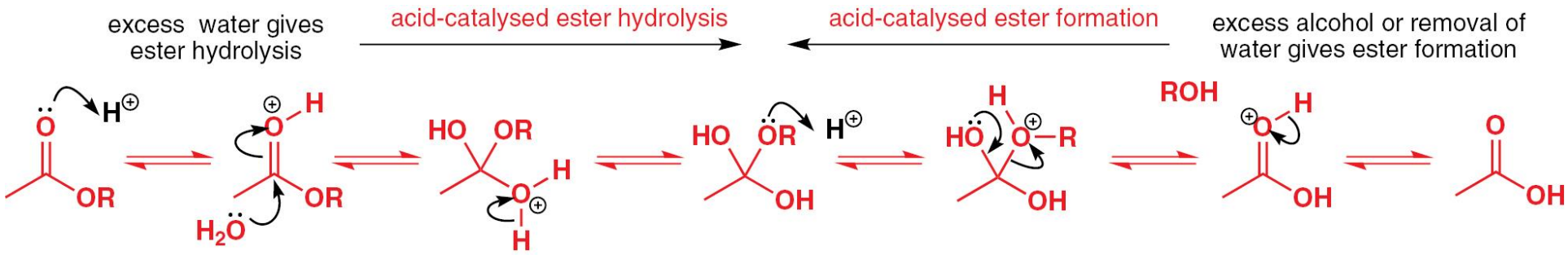
Acid-catalyzed ester hydrolysis and transesterification



Example



Acid-catalyzed ester hydrolysis and transesterification

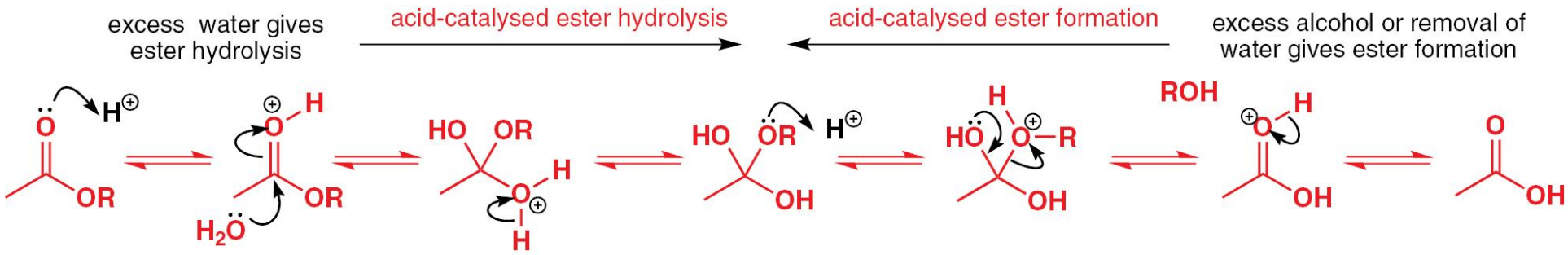


Example



Mechanism

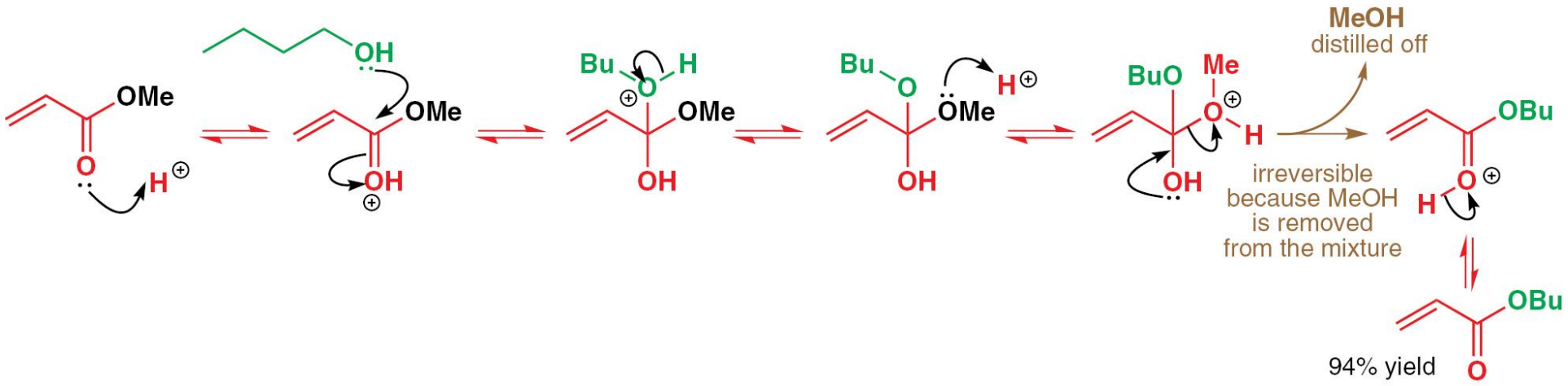
Acid-catalyzed ester hydrolysis and transesterification



Example



Mechanism



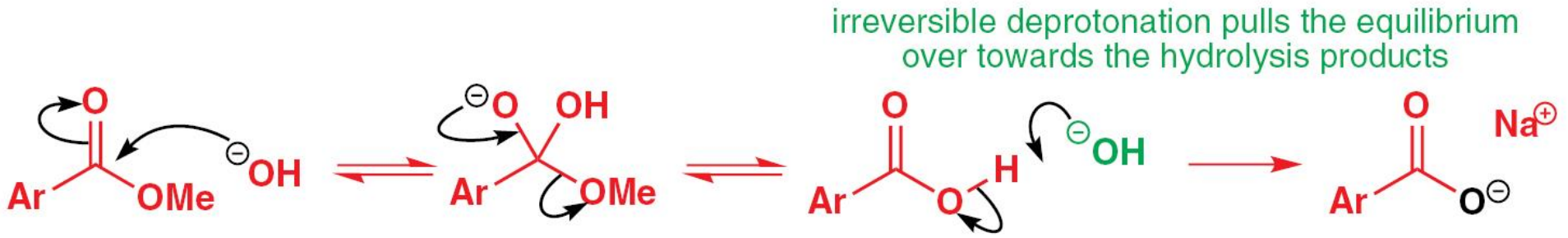
Base-catalyzed ester hydrolysis is irreversible



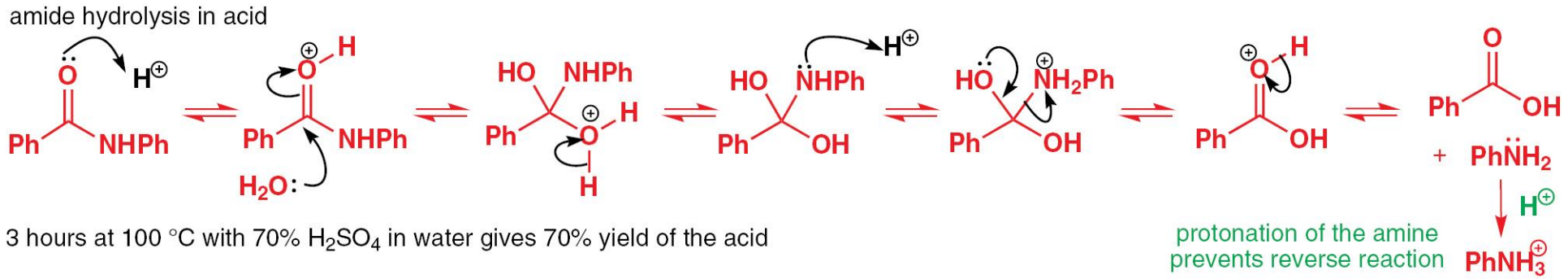
Base-catalyzed ester hydrolysis is irreversible



Mechanism

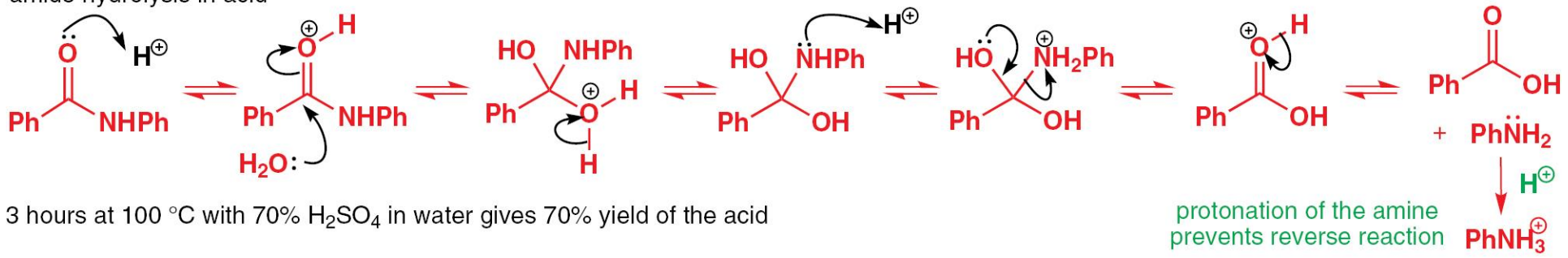


Acid and base-catalyzed amide hydrolysis



Acid and base-catalyzed amide hydrolysis

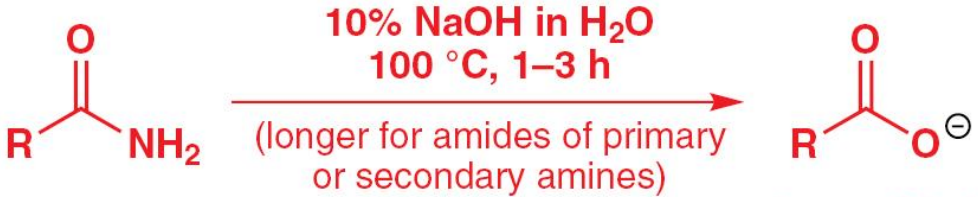
amide hydrolysis in acid



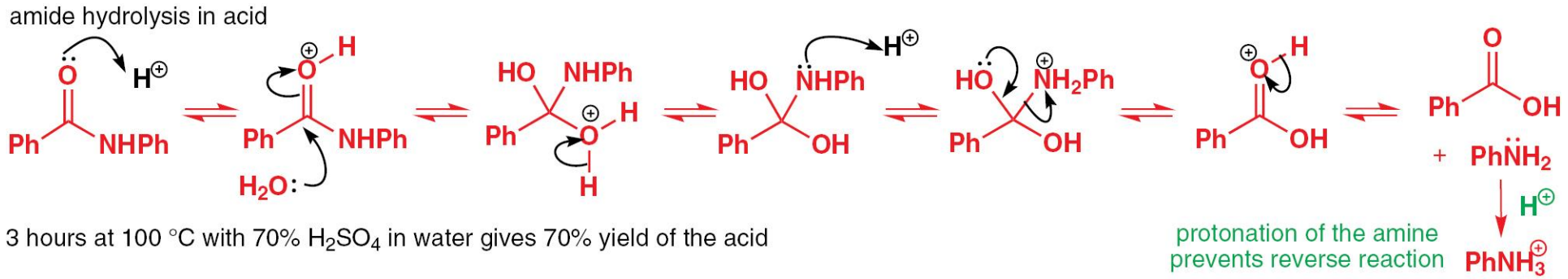
3 hours at 100 °C with 70% H_2SO_4 in water gives 70% yield of the acid

protonation of the amine prevents reverse reaction

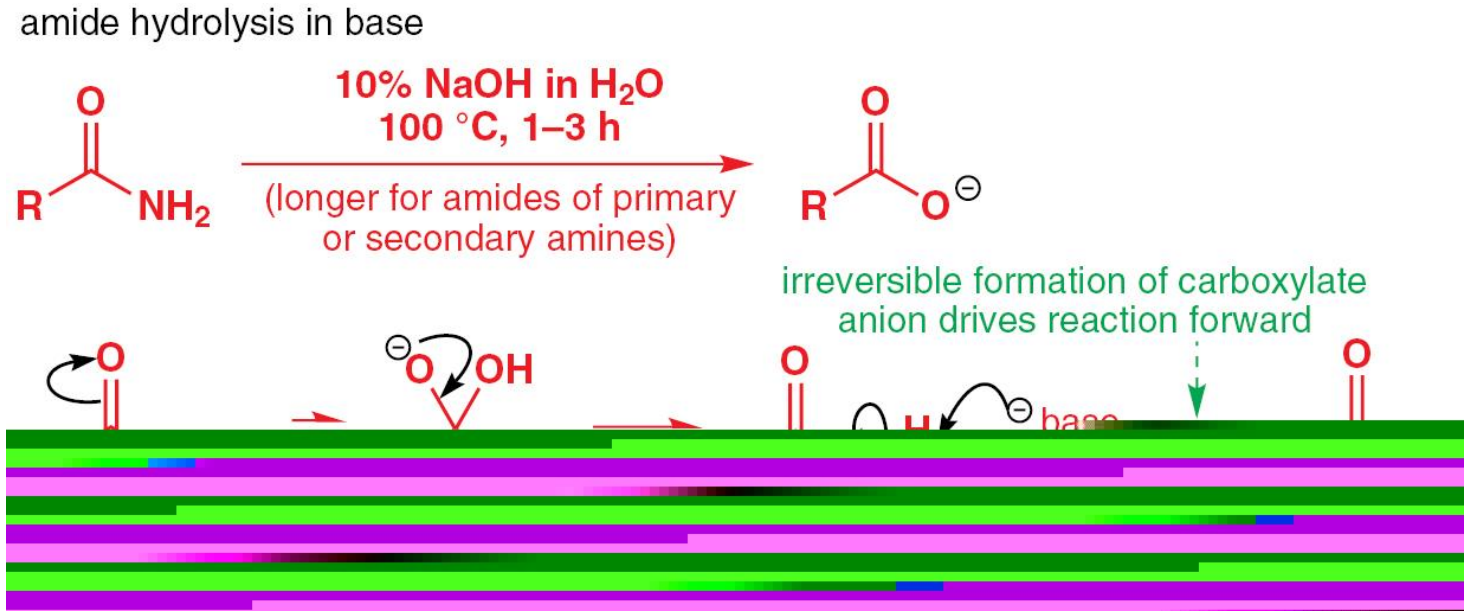
amide hydrolysis in base



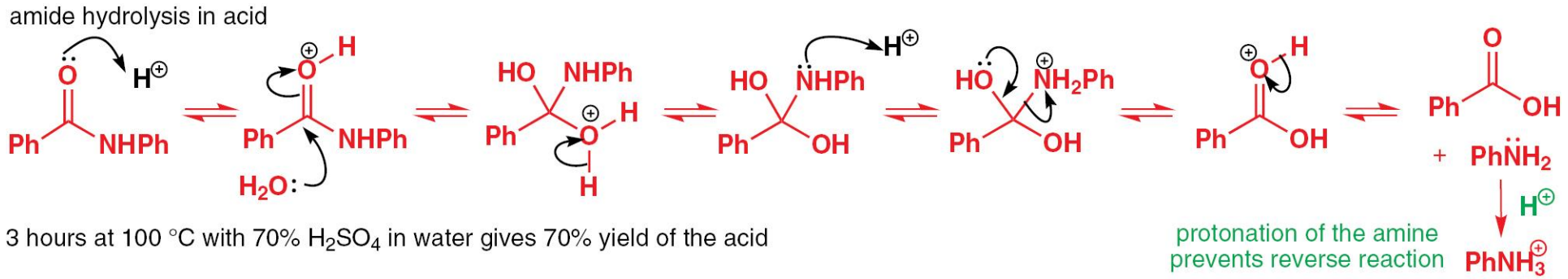
Acid and base-catalyzed amide hydrolysis



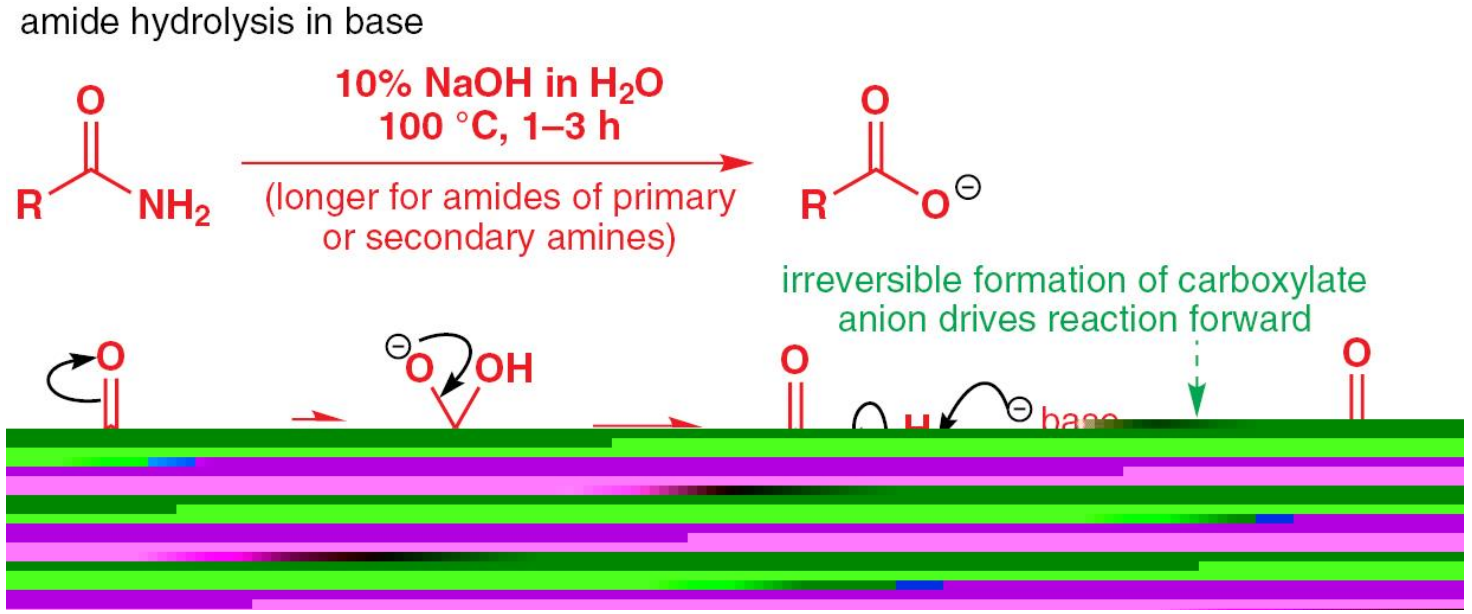
Mechanism



Acid and base-catalyzed amide hydrolysis



Mechanism

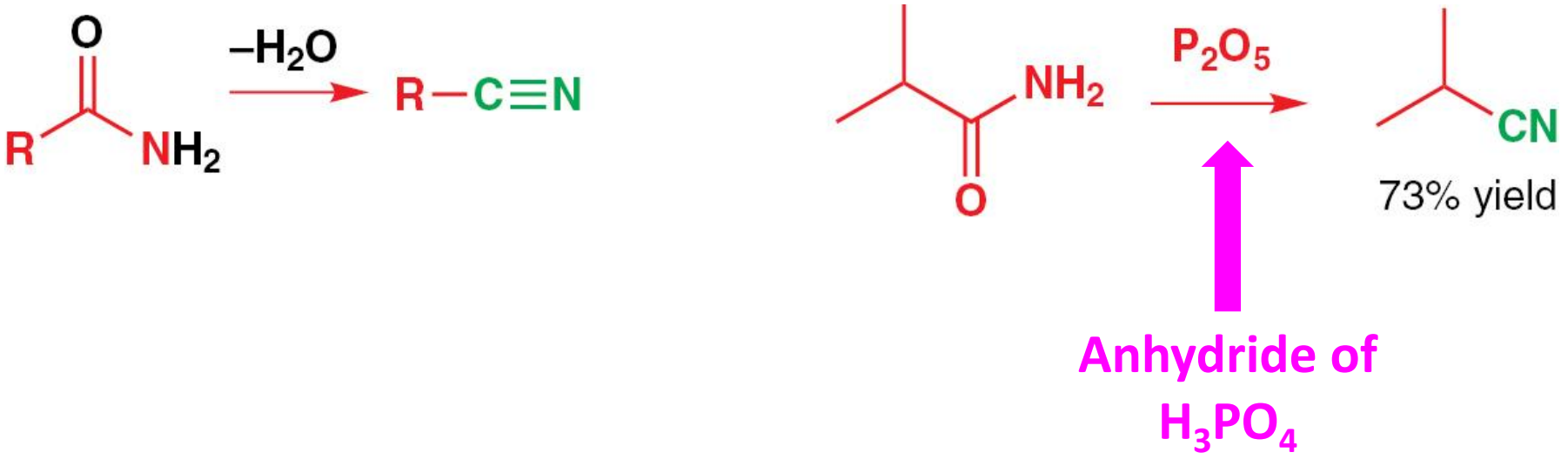


H_2O p*K*a=15.7, NH_3 p*K*a=35

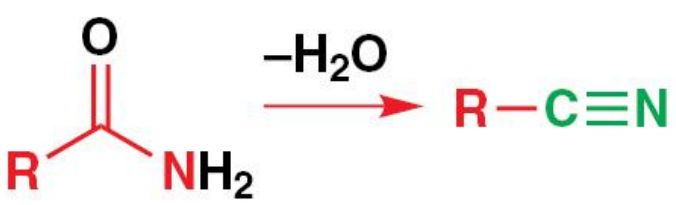
Amide dehydration-Nitrile hydrolysis

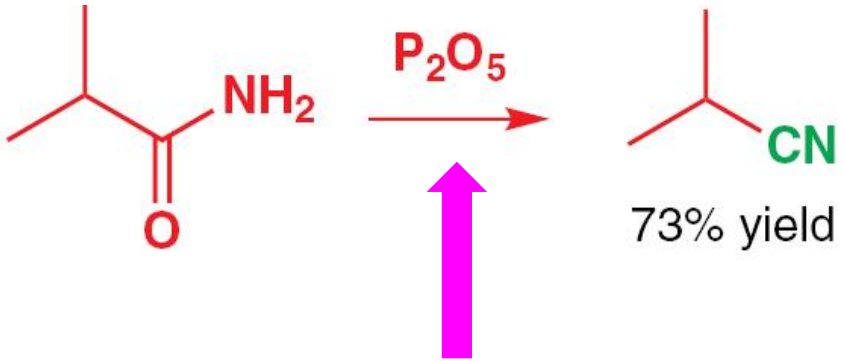


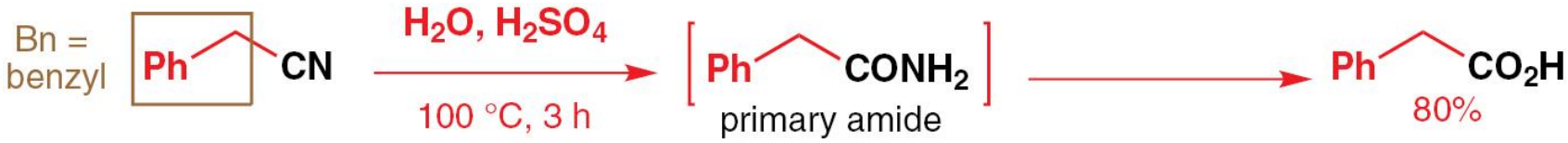
Amide dehydration-Nitrile hydrolysis



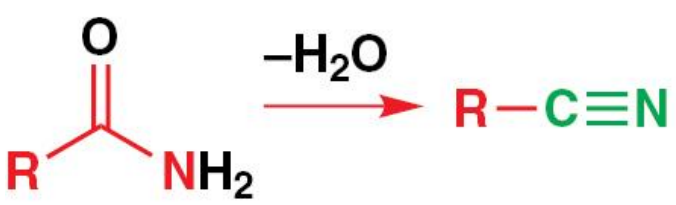
Amide dehydration-Nitrile hydrolysis

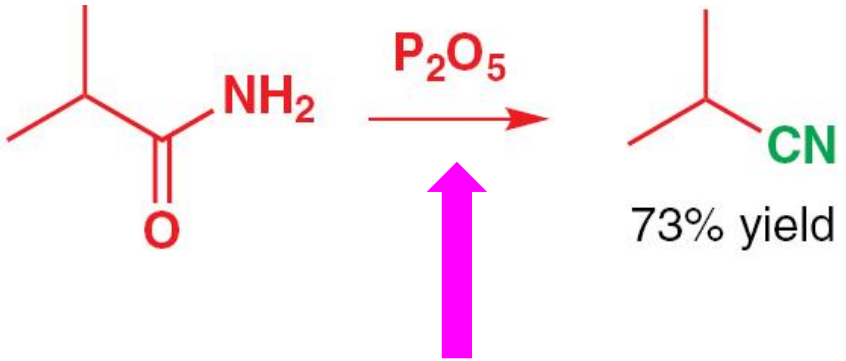


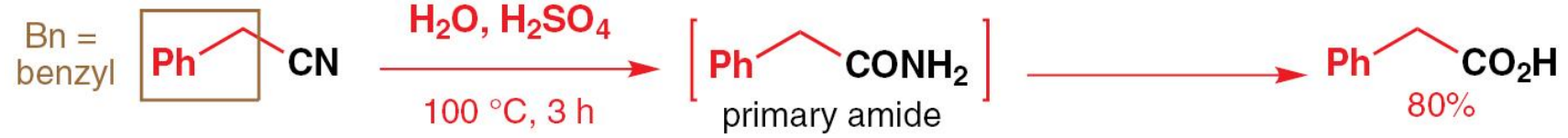


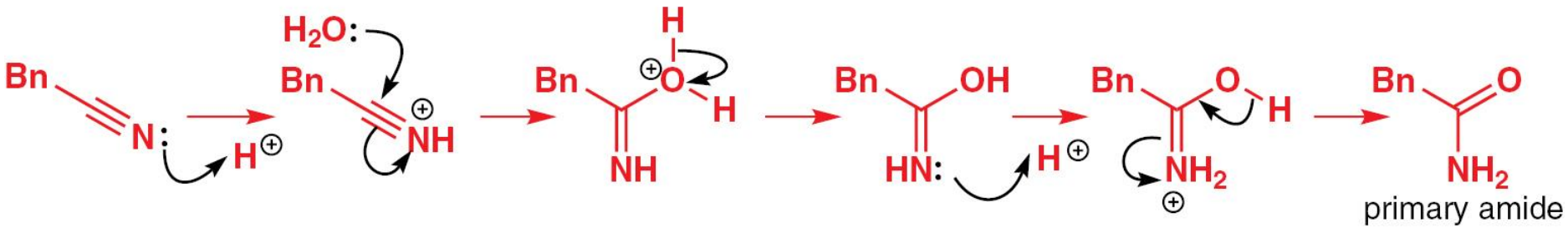


Amide dehydration-Nitrile hydrolysis

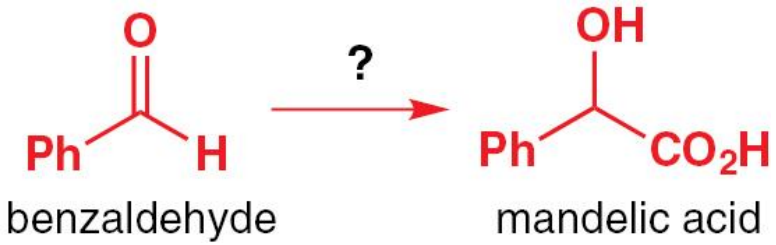








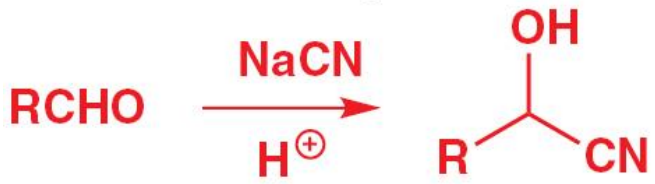
How could you synthesize mandelic acid?

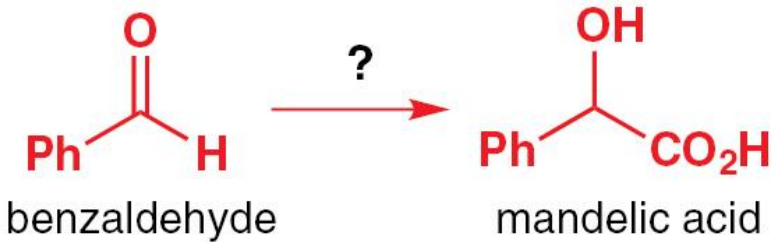


benzaldehyde mandelic acid

How could you synthesize mandelic acid?

reminder from Chapter 6:

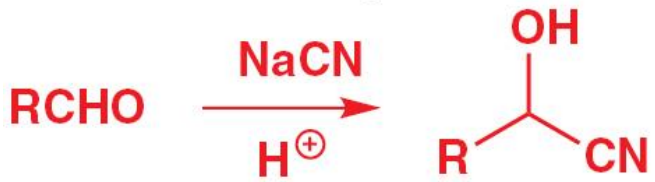


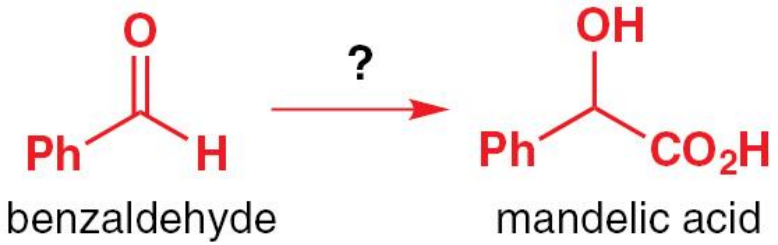


benzaldehyde mandelic acid

How could you synthesize mandelic acid?

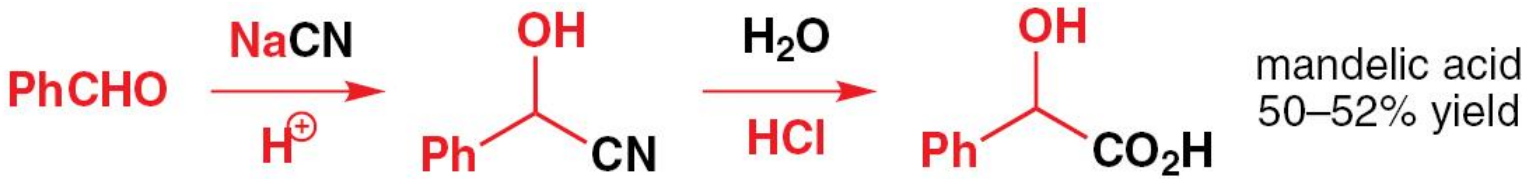
reminder from Chapter 6:





benzaldehyde mandelic acid

synthesis of mandelic acid from benzaldehyde



mandelic acid
50–52% yield